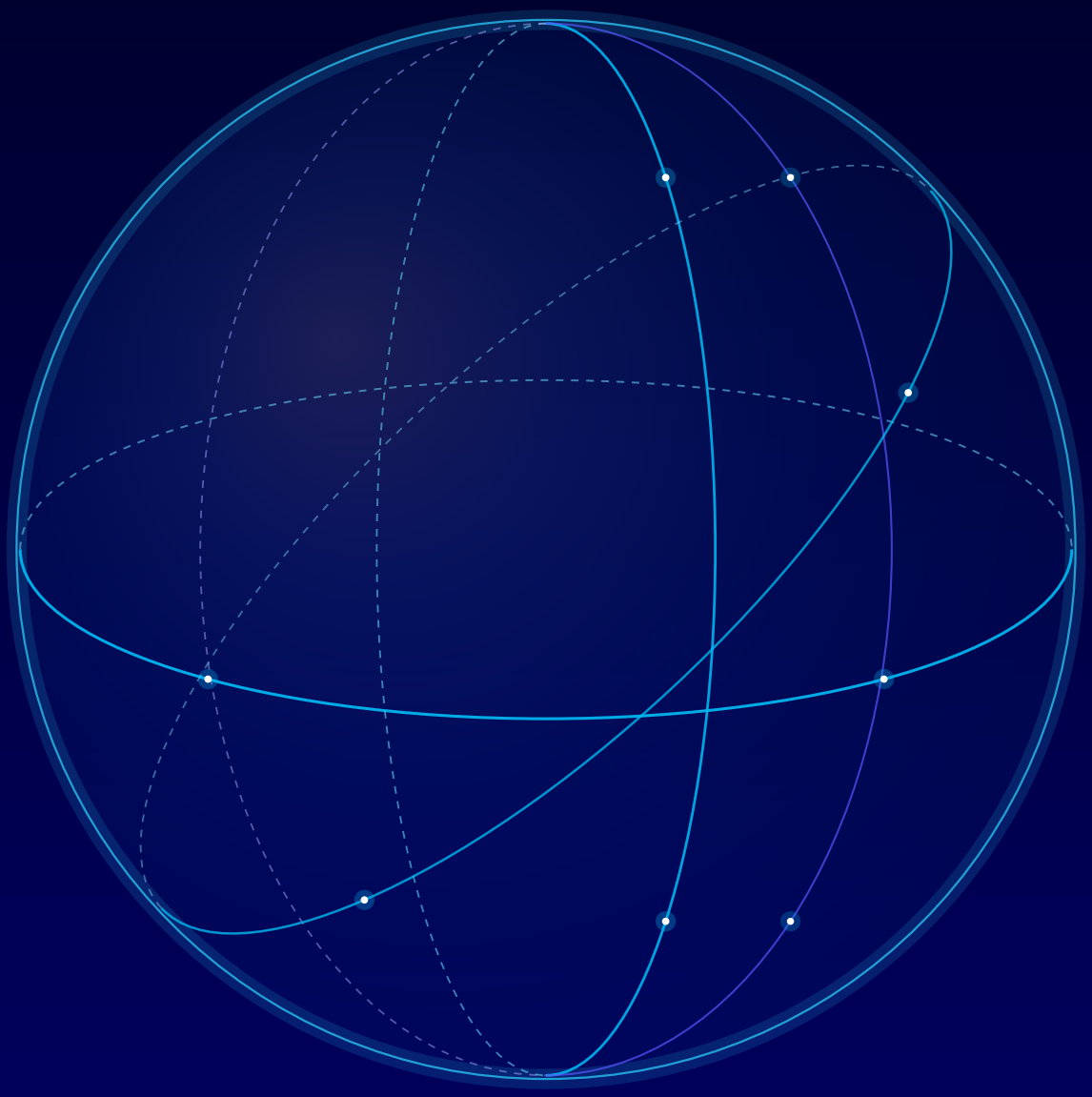
spherical astronomy, coordinates, and time



vaibhav

fizicyst

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Prerequisite Knowledge

- Basic trig identities (the angle sum and difference ones, you know the ones)
- Basic vectors: dot and cross products, and what they actually mean geometrically
- Basic differentiation and integration (chain rule, related rates)
- Zero regard for your schedule

1 Introduction

Okay, picture two astronomers. One's in Manila, one's in Manchester, and they're both staring at the same star at the exact same moment. You ask each of them where it is and they both go "uh, high up and kinda to the left." Cool, except they're pointing in totally different directions in space, because their "up" and "left" depend on where they happen to be standing on a spinning ball. So the very first problem in celestial navigation isn't even about stars. It's about getting everybody to agree on how to say where something is. That's all a coordinate system is: a rulebook so two people, anywhere on Earth and at any time, can point at the same spot in the sky and say the same thing about it.

2 Definition of Terms

- **Celestial sphere**: an imaginary sphere centered on the Earth with an infinite radius. Yeah, it sounds silly, but it's honestly a super handy trick. Stars are so ridiculously far away that their different distances don't matter for pointing, so we just pretend they're all painted on the inside of one giant dome and only care about *direction*, never distance.
- **Zenith (Z)**: the point on the celestial sphere straight above your head.
- **Nadir**: the point straight below your feet, 180° from the zenith.
- **Latitude**: goes from 0° at the equator to 90° at either pole. Every point that shares a latitude sits on a **parallel of latitude**. The famous ones are the tropic of Cancer ($23.5°N$), the equator (0°), the Arctic Circle ($66.5°N$), the Antarctic Circle ($66.5°S$), and the tropic of Capricorn ($23.5°S$).
- **Longitude**: goes from 0 to 180° east or west of the prime (Greenwich) meridian.
- **Great circle**: a circle on a sphere's surface that shares the sphere's center, so it's as big as the sphere itself. Basically it's the "straightest possible path" you can walk when you're stuck on the surface of a sphere.
- **Local meridian**: the great circle on your sky that goes from the north point of the horizon, up over the zenith, and down to the south point. Every longitude on Earth gets its own local meridian.

3 Horizontal and Equatorial Coordinates

3.1 Horizontal Coordinate System (a.k.a Alt-Az)

So the most natural way to say where a star is also happens to be the most personal one: how high is it above *your* horizon, and which way do you turn to face it? That's the **horizontal (Alt-Az)** system.

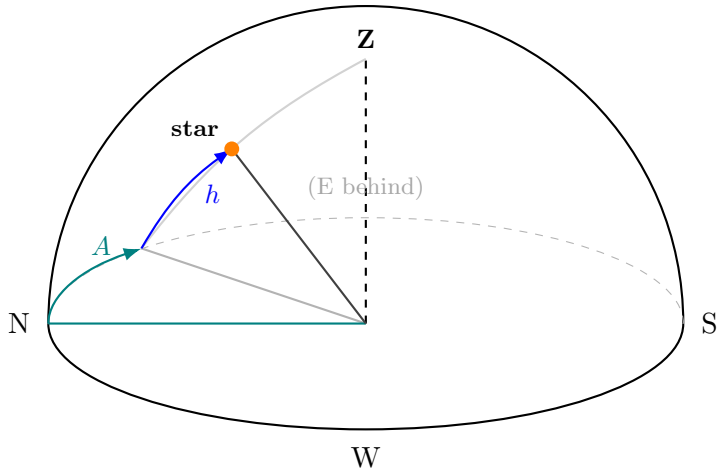


*Figure 1: Your personal sky bubble. The **altitude** h is how high the star is above the horizon, from 0 at the horizon up to 90° at the zenith Z. The **azimuth** A is how far around the horizon you turn, starting at north and going east, from 0 to 360°.*

Here's the catch though. Horizontal coordinates belong entirely to *you*. Move a few miles, or just wait around while the Earth rotates, and both h and A change. Two people looking at the same star at the same moment will give totally different Alt-Az numbers. So it's perfect for aiming a telescope and basically useless for making a star catalogue.

One thing you probably wanna remember: if you stand right at a pole, the celestial equator runs along your horizon. If you stand on the equator, the north celestial pole sits right on your north horizon and the south celestial pole sits right on your south horizon.

3.2 Equatorial Coordinate System (a.k.a RA and Dec)

Okay, so what if we wanna catalogue a star once and have it work for everyone, everywhere? Then we need a system that doesn't care where you're standing. It has to be stuck to the sky itself. That's the **equatorial** system.

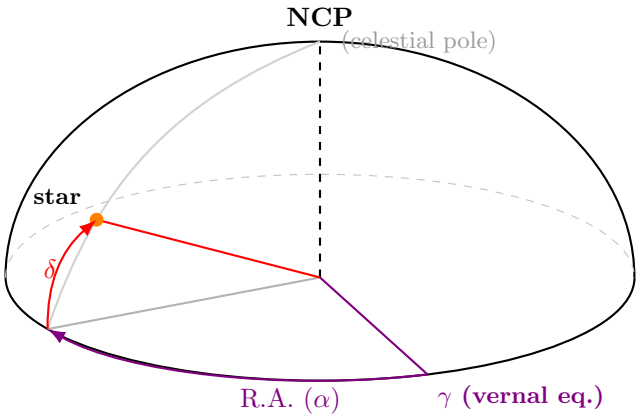


*Figure 2: The equatorial system, viewed from "above" the pole. **Declination** δ is the star's angular position north or south of the equator, ranging from -90 + 90 degrees, so basically a celestial latitude. **Right ascension** (R.A., α) is measured eastward from a fixed reference point on the celestial equator called the **vernal equinox** (γ). Think of it as celestial longitude, except it*

never turns with the observer.

Declination is just the sky's version of latitude, and since stars are so ridiculously far away, their declinations stay basically fixed for decades. The Sun is the big exception. It's way closer, and Earth's axis is tilted 23.5° relative to its orbit, so the Sun's declination swings from $23.5°S$ up to $23.5°N$ and back over a year. That swing is the whole reason we get seasons.

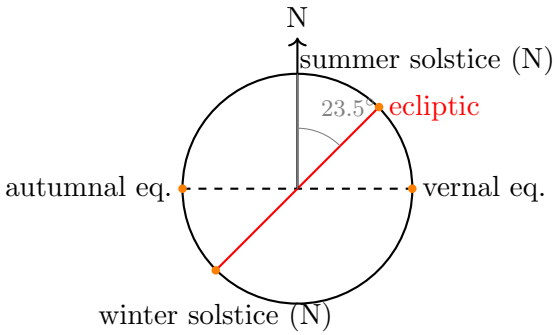


Figure 3: The Sun's declination over the year, drawn as the tilt of the ecliptic (the Sun's apparent path) against the celestial equator. At the summer solstice the Sun sits at its northernmost declination (+23.5°, straight over the tropic of Cancer). At the winter solstice it's at its southernmost (−23.5°, over the tropic of Capricorn). Flip the labels for the southern hemisphere. At either equinox the Sun crosses the celestial equator, so $\delta = 0°$.

Right ascension is the celestial longitude, but it's pinned to the vernal equinox instead of Greenwich, and we usually write it in *hours* instead of degrees. A full circle is 360° and there are 24 hours around it, so every 15° of R.A. is 1 hour. So an R.A. of 20° is 1^h20^m.

3.3 Synthesis of the Horizontal and Equatorial Coordinate Systems

Quick recap, no stress. Horizontal coordinates (h, A) are all about the *observer*, change constantly, and are what you'd punch into a telescope mount. Equatorial coordinates (δ, α) are all about the *sky*, stay put for a star over human timescales, and are what you'd find in a star atlas. Neither one is "better." They just answer different questions, and the next chunk of this handout is about building a bridge between them.

4 Hour Angle (H)

To build that bridge we need one more ingredient: how far has a star traveled across the sky since it last crossed your local meridian? That's the **hour angle**, H. It's the angle between two meridians (the star's and your local one), measured to the west of your local meridian.[1]

[1]Some books measure hour angle to the east instead. Either way works, just don't mix them up.

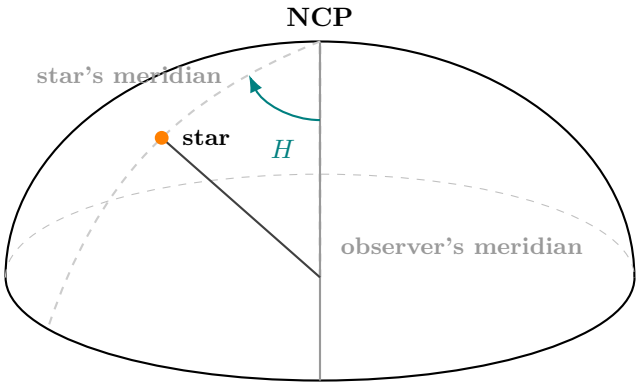


Figure 4: Hour angle H. It's zero when the star is right on your local meridian, then grows as the star drifts west. We call west "forward in time" because stars rise in the east and set in the west.

5 Sidereal Time

The Earth spins once on its axis every $23^h56^m4^s$. That's a **sidereal day**, measured against the fixed background stars. It's a few minutes shorter than the 24 hour solar day you're used to, because during that day the Earth also moved a bit along its orbit, so it has to "catch up" a little extra before the Sun is back where it started. During one sidereal day, every right ascension value on the celestial sphere sweeps past your local meridian exactly once. Whatever R.A. is on your local meridian right now is your **local sidereal time (LST)**.

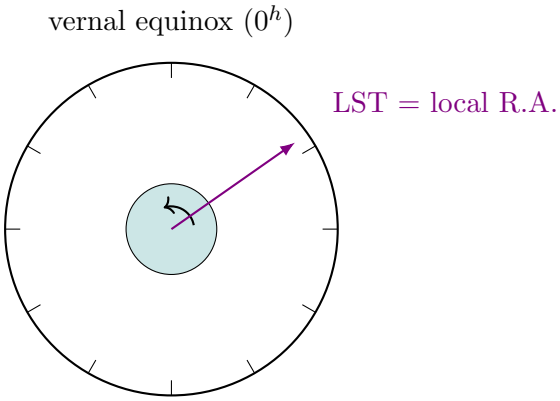


Figure 5: The R.A. "clock face" stays fixed against the stars while the Earth (center) turns underneath it. The arrow points at whichever R.A. is on the observer's local meridian right now, and that's the LST.

Here's a neat trick. Anything sitting on your local meridian has an hour angle of zero, by definition. And a star's hour angle plus its right ascension has to add up to the meridian's total sweep. Put those two facts together and you get something super simple and super useful:

$$\boxed{LST = H + \alpha.}$$

So if you know the right ascension of any star you can see, plus its hour angle right now, you instantly know your local sidereal time.

6 Circumpolarity

6.1 Upper and Lower Culminations

A star hits **upper culmination** when it crosses your local meridian at its *highest* point of the day, and **lower culmination** when it crosses at its *lowest*. A star that rises and sets only ever shows you its upper culmination (the lower one happens below the horizon). A star that never dips below the horizon crosses the meridian twice a day, once high and once low, and we call that **circumpolar**. (We're using the convention that the northern hemisphere is positive.)

6.2 Maximum and Minimum Altitudes

At upper culmination, the star, the NCP, and the zenith are all on the same meridian, so the max altitude is just 90° minus the absolute difference between declination and latitude:

$$\boxed{h_u = 90^\circ - |\phi - \delta|.}$$

At lower culmination it's the same idea, except now the star's on the *other* side of the pole from the zenith, and you get the min altitude of a circumpolar object:

$$\boxed{h_l = |\phi + \delta| - 90^\circ.}$$

Fun fact: these two equations are secretly a linear system in ϕ and δ. Subtract them and the declination cancels, so you're left with just latitude:

$$h_u - h_l = 180^\circ - 2\phi \quad \Longrightarrow \quad \boxed{\phi = \frac{h_l - h_u + 180^\circ}{2}.}$$

Add them instead and the latitude cancels:

$$\boxed{\delta = \frac{h_u + h_l}{2}.}$$

(If you ever get an angle bigger than 90°, like 100° south, take its difference from 90° (that's 10°) and subtract that from 90° again. So 100° south turns into 80° south.)

6.3 Declination of Non-Rising and Non-Setting Stars

A star never sets (so it's circumpolar) at latitude ϕ exactly when its minimum altitude is positive, $h_l > 0$:

$$\boxed{|\phi + \delta| > 90^\circ.}$$

Same logic the other way: a star never rises at all when its *maximum* altitude is negative, $h_u < 0$:

$$\boxed{|\phi - \delta| > 90^\circ.}$$

7 Sun Rising and Set Times

There's one neat formula that gives you the time of sunrise or sunset:

$$\boxed{\cos H = -\tan\phi \tan\delta,}$$

where δ is the Sun's declination and ϕ is your latitude. We'll actually derive it once we've got spherical trig (it's in Section 12.1.3), so for now just trust me. Turn the H you get into hours (every 360° is 24^h), pretend sunrise and sunset are symmetric around local noon (we're ignoring how much the Sun's declination drifts during one day), and boom, you've got rise and set times.

8 The Horizon at Heights

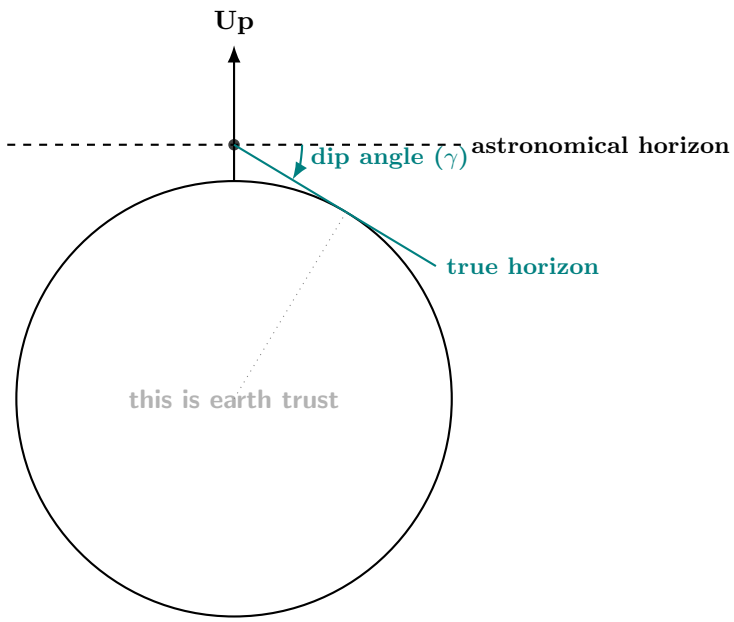


Figure 8: The dip angle. If you're at a height h above sea level (the gray dot, floating above the Earth's surface), your true horizon, meaning the line of sight that just skims the Earth's curve, dips below the eye-level astronomical horizon.

Climb a mountain and you can see further below eye level than you could at sea level. That extra angle you gain from being at height h is the **dip angle**:

$$\boxed{\Theta_{dip} = \cos^{-1}\left(\frac{R}{R+h}\right),}$$

where R is Earth's radius. It's literally just a right triangle made of your height above the ground and the tangent line to the surface.

9 The Ecliptic

The **ecliptic** is the great circle the Sun traces across the celestial sphere over a year. Another way to say it: it's Earth's orbital plane projected onto the sky. As the Sun goes around it, it passes in front of a bunch of constellations, the **zodiac constellations**.

Zodiac Constellation	Dates
Capricornus	Jan 20 to Feb 16
Aquarius	Feb 16 to Mar 11
Pisces	Mar 11 to Apr 18
Aries	Apr 18 to May 13
Taurus	May 13 to June 21
Gemini	June 21 to July 20
Cancer	July 20 to Aug 10
Leo	Aug 10 to Sep 16
Virgo	Sep 16 to Oct 30
Libra	Oct 30 to Nov 23
Scorpius	Nov 23 to Nov 29
Ophiuchus	Nov 29 to Dec 17
Sagittarius	Dec 17 to Jan 20

Fun thing: the Sun actually spends about 18 days a year passing through Ophiuchus, which is more time than it spends in Scorpius (just six days in the table above). But the ecliptic only barely clips the corner of Ophiuchus on a sky map, so it just gets left off the list.

10 Spherical Triangle

10.1 Definition of a Spherical Triangle

So far it's all been definitions and one-off formulas. To actually *derive* how horizontal and equatorial coordinates connect, we need a proper tool: the **spherical triangle**.

Take three great circles on a sphere. Every pair crosses at two (antipodal) points, and if you pick three of those crossing points smartly, the arcs between them wrap around a triangle that lives entirely on the sphere's curved surface. That's a spherical triangle, and no, it is *not* the flat triangle from your geometry class. Its three "sides" are arcs of great circles that curve along the sphere, and its angles (the angles between the *planes* of the great circles at each vertex) don't have to add up to 180°. Spoiler: as we'll see in Section 11.4, they always add up to *more*.

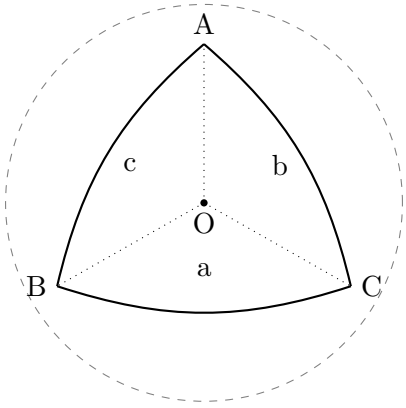


Figure 9: A spherical triangle. The dashed circle is the sphere, the solid curved arcs a, b, c are great circle arcs on its surface, and O is the center. Lowercase letters are arc lengths (same thing as the central angles, so the arc $a = \angle BOC$), and uppercase letters are the vertex angles. For example, $\angle A$ is the angle between the planes AOB and AOC.

10.2 Visualizing the Relationship between Eqr-Hor Coordinate Systems using ST

Okay, let's put this thing to work. Let the sphere be the celestial sphere, and put the three vertices at your zenith Z, the north celestial pole N, and some object X in the sky.

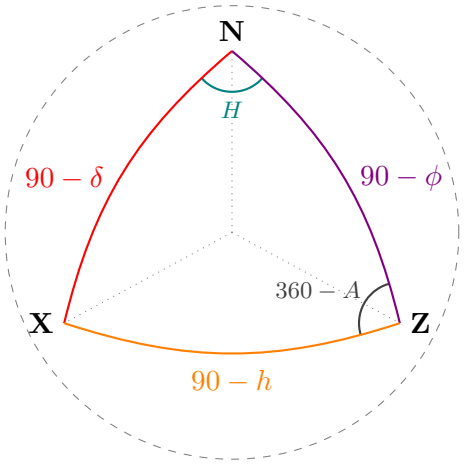


Figure 10: The spherical triangle NXZ, with the north celestial pole, a celestial object, and the observer's zenith at its corners. This one triangle is the whole bridge between horizontal and equatorial coordinates.

Every side of this triangle is something we already know. The arc $\widehat{ZX}$ is the angle between the zenith and the object, which is the complement of altitude: $\widehat{ZX} = 90 - h$. The arc $\widehat{ZN}$ is the angle between the zenith and the pole, which is $90 - \phi$. And $\widehat{XN}$ is the complement of declination, $90 - \delta$.

What about the angles at the corners? At the pole N, the angle between the object's meridian and your meridian is, by definition, the hour angle: $\angle N = H$. At the zenith Z it works out to 360° minus the azimuth. That's because azimuth increases *eastward*, but the angle in our triangle goes the other way around:

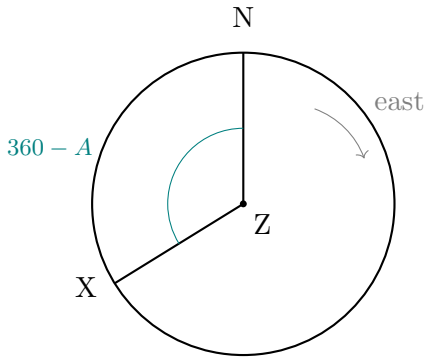


Figure 11: A flat top down view of the observer's horizon, just to nail down the azimuth

convention. Azimuth increases eastward, but the triangle's angle at Z opens the other way, so $\angle Z = 360 - A$.

So now we've got every side and every angle of NXZ written in terms of stuff we already understand. All that's left is to throw this triangle into the spherical trig machine, which is exactly what Section 12 does.

11 Spherical Trigonometry

11.1 Law of Cosines

The claim: for a spherical triangle with sides a, b, c and opposite angles A, B, C,

$$\boxed{\cos a = \cos b \cos c + \sin b \sin c \cos A.}$$

Why it's true (sketch): stick the sphere's center at the origin of regular 3D space with radius 1, and put vertex A at the north pole $(0, 0, 1)$. Vertices B and C then sit somewhere else on the unit sphere, and we can write their coordinates with ordinary spherical coordinates (an angle around the pole and an angle down from it). Now find the straight line (chord) distance between B and C two different ways: once with the usual flat space distance formula, and once with the flat triangle law of cosines on the isosceles triangle OBC (two sides of length 1, included angle a). Set the two answers equal, grind through the algebra (a term $\cos(\phi_B - \phi_C)$ shows up, which the angle difference identity $\cos(x-y) = \cos x \cos y + \sin x \sin y$ unpacks), and the boxed formula pops out. The algebra is boring and mechanical, so I'm not gonna write out every line. What's worth keeping is that the spherical law of cosines is really just ordinary 3D Pythagorean style geometry written in angles instead of lengths.

11.2 Law of Sines

The claim:

$$\boxed{\frac{\sin A}{\sin a} = \frac{\sin B}{\sin b} = \frac{\sin C}{\sin c}.}$$

The proof needs more vector machinery than we want to deal with here, but the formula itself is easy to use. Any time you know two angles and a side (or two sides and an angle) that sit opposite each other, this one line solves the triangle.

11.3 Napier's Circle

A **right spherical triangle** is one with a 90° corner. That special case gives you a bunch of shortcut formulas, and there's a nice trick to remember all of them at once: **Napier's circle**.

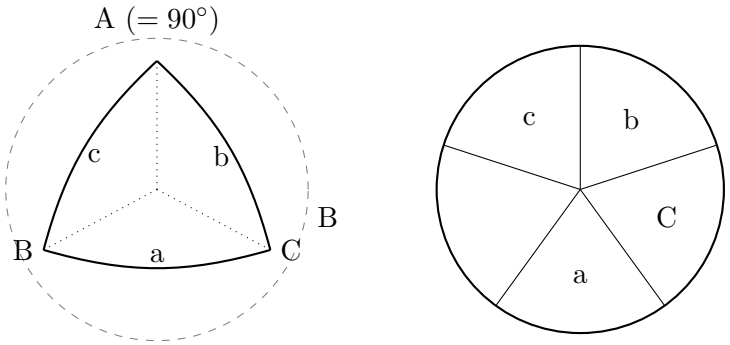


Figure 12: Left, a right spherical triangle, with the curved sides drawn just like Figure 9 but with one corner (A) stuck at 90°. Right, Napier's circle: the five other *elements of the triangle (everything except the known 90° angle) laid out around a wheel in the same order they show up going around the triangle.*

One catch before you read anything off the wheel: swap the three elements that are *not* legs (the hypotenuse a and the two angles B and C) for their complements, $90° - a$, $90° - B$, and $90° - C$. The legs b and c stay as they are.

Two rules fall out of this wheel:

Tan-ad rule: the sine of any element equals the product of the tangents of the two elements *adjacent* to it on the wheel. For the element $90° - B$, the neighbors are c and $90° - a$, so

$$\boxed{\cos B = \tan c \cot a.}$$

Co-op rule: the sine of any element equals the product of the cosines of the two elements *opposite* to it (the non-adjacent ones). For the element $90° - B$, the opposite elements are b and $90° - C$, so

$$\boxed{\cos B = \cos b \sin C.}$$

($90° - B$ can be swapped for any of the five elements, just be careful about who its adjacent and opposite neighbors are.) You don't need to memorize either formula. Just know how to draw the wheel and read the relationship right off it.

11.4 Area of a Spherical Triangle

We already dropped the hint that a spherical triangle's angles add up to more than 180°. The excess, $E = A + B + C - 180°$, is way more than a curiosity. It's directly proportional to the triangle's area:

$$\boxed{A_{area} = ER^2,}$$

where E has to be in radians. This makes total sense at the extremes: if a spherical triangle's angles added up to exactly 180°, the formula says its area would be exactly zero. So any spherical triangle with real, nonzero area must have some actual excess above 180°, which is just a fancy way of saying its sides have to be visibly curved, not flat.

12 Applications of Spherical Trigonometry in Astronomy

12.1 Relationships between Eqr and Hor Coordinate System

Time to cash in that NXZ triangle from Section 10.2.

12.1.1 Solving for the Altitude from Equatorial Coordinates

Use the spherical law of cosines on arc $\widehat{ZX}$ (it's opposite vertex N, the included angle is H, and the adjacent sides are $\widehat{ZN} = 90 - \phi$ and $\widehat{XN} = 90 - \delta$):

$$\cos(90 - h) = \cos(90 - \delta)\cos(90 - \phi) + \sin(90 - \delta)\sin(90 - \phi)\cos H.$$

Use $\cos(90 - x) = \sin x$ and $\sin(90 - x) = \cos x$ everywhere and it turns into

$$\boxed{\sin h = \sin\delta \sin\phi + \cos\delta \cos\phi \cos H.}$$

One equation, and it ties together all four things we've been working toward: altitude, declination, latitude, and hour angle.

12.1.2 Solving for the Declination from Horizontal Coordinates

Do the same law of cosines trick on the side $\widehat{XN} = 90 - \delta$ instead (it's opposite vertex Z, the included angle is $360 - A$, and remember $\cos(360 - A) = \cos A$):

$$\boxed{\sin\delta = \sin h \sin\phi + \cos h \cos\phi \cos A.}$$

12.1.3 Hour Angle Formula (Sunrise/Sunset)

Set $h = 0$ (the object is sitting right on the horizon) in the altitude formula and you get back the sunrise/sunset relation I promised you in Section 7:

$$0 = \sin\delta \sin\phi + \cos\delta \cos\phi \cos H \quad \Longrightarrow \quad \boxed{\cos H_\odot = -\tan\delta_\odot \tan\phi.}$$

Convert H to hours using $t(\text{hr}) = \frac{H}{360^\circ} \cdot 24^h$. Then sunset is at $12{:}00 + \frac{H_\odot}{360^\circ} \cdot 24^h$ and sunrise is at $12{:}00 - \frac{H_\odot}{360^\circ} \cdot 24^h$ (assuming a circular orbit and a constant solar declination over the course of one day).

12.2 Local Sidereal Time

Notice something? Right ascension α never showed up in Sections 12.1.1 to 12.1.3, even though it's one of our four horizontal and equatorial parameters (h, A, δ, α). We don't need those formulas to bring α in, because we already built the connector back in Section 5:

$$\boxed{LST_{observer} = H_{object} + \alpha_{object}.}$$

And if you solve for the Sun's own hour angle and right ascension, you get an even more useful version:

$$\boxed{LST = H_\odot + \alpha_\odot = H_{object} + \alpha_{object}.}$$

12.3 Local Time

Local time (LT), also called **mean solar time**, is clock time tied to the Sun's hour angle, assuming the Sun moves across the sky at a perfectly constant rate all year long (we're ignoring the real eccentricity of Earth's orbit and the tilt of the ecliptic, more on that in Section 14.4):

$$\boxed{LT = H_{\odot,\text{hours}} + 12^h.}$$

12.4 Extra Horizon Angle

We already saw the dip angle in Section 8. Here's the more careful version, with x for height and $R_\oplus$ for Earth's radius:

$$\boxed{\cos\Theta = \frac{R_\oplus}{R_\oplus + x}.}$$

At a given height x, sunrise and sunset genuinely shift, since the Sun has to clear this extra angle Θ before it hits your *true* (raised) horizon instead of the eye level (astronomical) one. There's also **atmospheric refraction**, which is just light bending as it goes through the atmosphere. It lifts things near the horizon by about $34'$, and we usually ignore it unless a problem tells us not to.

12.5 Sample Problems

Problem: Resolving Mizar and Alcor

A telescope's minimum angular resolution is $\Theta = 1.22\lambda/D$. Mizar ($\alpha = 13^h26^m16^s$, $\delta = 54°51'33.2''$) and Alcor ($\alpha = 13^h24^m58.3^s$, $\delta = 54°47'46.5''$) appear as a single point to the naked eye. What telescope diameter would be needed to resolve them, using $\lambda = 550\,nm$?

Convert both coordinates to decimal degrees and use the spherical law of cosines to get the angle Θ between the two stars:

$$\cos\Theta = \cos(90-\delta_A)\cos(90-\delta_M) + \sin(90-\delta_A)\sin(90-\delta_M)\cos(\alpha_A - \alpha_M)$$

That gives $\Theta \approx 0.1968° = 3.4355\times10^{-3}$ rad. Plug into $D = 1.22\lambda/\Theta$ and you get $D_{min} \approx 0.195\,mm$. That's way smaller than a human pupil (2 to $8\,mm$), so most people can split these two with the naked eye if they really focus.

Problem: Altitude of Eltanin from Quezon City

Eltanin sits at $\delta = 51.5°N$. One night it transits the local meridian of Quezon City ($\phi = 14.5°N$). Find its altitude.

At transit $H = 0$, so the altitude formula just collapses to $\sin h = \sin\delta\sin\phi + \cos\delta\cos\phi$, which is literally $\cos(\phi - \delta)$. Plug in the numbers and you get $h = 53°$ (so a zenith distance of $37°$, its complement).

13 Spherical Trigonometric Relationships between Equatorial and Ecliptic Coordinates

Okay, we've linked horizontal and equatorial. Let's add a third system: **ecliptic coordinates**, which you get by tilting the whole equatorial system by the axial tilt $\epsilon = 23.5°$. A point now has an **ecliptic latitude** β and an **ecliptic longitude** λ (increasing eastward), and we call the pole of this system K, the ecliptic north pole, so it doesn't get mixed up with the regular celestial pole N.

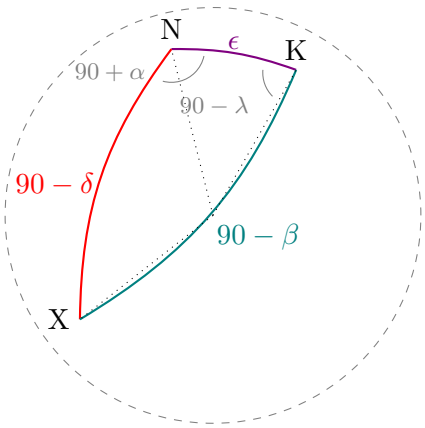


Figure 13: The spherical triangle with the celestial pole N, the ecliptic pole K, and an object X at its corners. The tilt angle ϵ sits right between the two poles.

13.1 Calculating the Ecliptic Latitude from the Ecl-Eqr Coordinates

Apply the law of cosines to this triangle:

$$\boxed{\sin\beta = \cos\epsilon\sin\delta - \cos\delta\sin\epsilon\sin\alpha.}$$

13.2 Calculating the Declination from Equatorial-Ecliptic Coordinates

Run the same triangle the other way around:

$$\boxed{\sin\delta = \cos\epsilon\sin\beta + \cos\beta\sin\epsilon\sin\lambda.}$$

13.3 Emergent Formulas from the Ecliptic-Equatorial Coordinates

The Sun always sits exactly on the ecliptic, so its ecliptic latitude is always $\beta_\odot = 0$. Plug that into the declination formula above and you get the sine wave behind the seasons that we sketched back in Figure 3:

$$\boxed{\sin\delta_\odot = \sin\epsilon\sin\lambda_\odot.}$$

The Sun's ecliptic longitude goes up at a roughly constant rate through the year, so

$$\boxed{\lambda = \frac{n}{365.242}\cdot 360^\circ,}$$

where n is the number of days since the vernal equinox and 365.242 is the length of the tropical year. One more result, and you'll need stuff from Section 14.2 to derive it:

$$\boxed{\tan\alpha_\odot = \cos\epsilon\tan\lambda_\odot.}$$

13.4 Sample Problem

Problem: The Galactic Pole and Vega

The northern galactic pole sits at equatorial coordinates $(\alpha_G, \delta_G) = (12^h51^m, 27°08')$. Find its ecliptic latitude and longitude, then find the galactic longitude of Vega $(\alpha_v, \delta_v) =$

$(18^h36^m56.3^s, 38°47'1.3'')$, given the galactic center sits at $(17^h45^m40^s, -29°00'28.1'')$.

Plug (α_G, δ_G) into the two boxed formulas above (with $\epsilon = 23.5°$) and you get $\beta_G = +29°46'$ and $\lambda_G = 179°53'$. (The declination formula by itself can't tell $\lambda \approx 0.1°$ from $\lambda \approx 179.9°$. The relation $\cos\lambda\cos\beta = \cos\delta\cos\alpha$ from Section 14.2 is negative here, which picks the second one.) For Vega's galactic longitude we make a second spherical triangle out of the galactic pole G, the galactic center C, and Vega V. The arc $\widehat{GC} = 90°$ (the center is on the galactic plane, by definition), $\widehat{GN} = 90 - |\delta_G|$, and $\widehat{CV}$ comes from the regular law of cosines using the equatorial coordinates of C and V, which gives $\widehat{CV} = 68.839°$. One more law of cosines, this time on triangle GVC, gives the galactic longitude $\iota_V = 67.53°$.

14 Other Spherical Trigonometric Relationships

14.1 Four Parts Formula

One more relation for the toolbox. It's handy when you know an arc (a) and the two angles at its ends (B and C) and want the other arc c that meets a at vertex B, without doing two separate law of cosines steps:

$$\boxed{\cot c = \frac{\cos a \cos B + \sin B \cot C}{\sin a}.}$$

14.2 Rotational Coordinate Transformations

Everything in Sections 12 and 13 can also be derived a totally different way: rotate a 3D Cartesian coordinate system by the angle between two vertices of the triangle, and match up the coordinates before and after. Concretely, if A sits at the pole $(0, 0, 1)$ and B, C are two other vertices separated by arc α, then rotating the whole frame by α turns B's coordinates into C's, which gives you three equations at once (one per axis) instead of just one. It's more setup than the law of cosines, but you get *three* formulas for the price of one. Do this on the NXZ triangle (letting $A = N$, $B = Z$, $C = X$) and you get, with azimuth measured from north through east and H measured westward like everywhere else in this handout,

$$\boxed{\cos H \cos\delta = \sin h \cos\phi - \cos A \cos h \sin\phi,} \qquad \boxed{\sin H \cos\delta = -\sin A \cos h,}$$

and those sit next to the altitude formula we already got (Section 12.1.1), which is the third component. Do the same thing with the ecliptic triangle from Section 13 and you get two more relations to go with the ones we already have:

$$\boxed{\sin\lambda\cos\beta = \sin\delta\sin\epsilon + \cos\delta\cos\epsilon\sin\alpha,} \qquad \boxed{\cos\lambda\cos\beta = \cos\delta\cos\alpha.}$$

14.3 Proper Motion

Stars aren't painted on the celestial sphere forever. Even if you ignore Earth's rotation, precession, and orbit, real stars slowly drift across the sky, usually taking decades to move even one arcminute. That drift is called **proper motion**, and it splits into two pieces: motion along a declination circle and motion along a meridian.

The meridian piece is easy: $u_\delta = \Delta\delta/\Delta t$. The declination circle piece needs one extra thought. A circle of constant declination δ is smaller than the celestial equator by exactly a factor of $\cos\delta$

(same reason lines of longitude squish together near Earth's poles). So

$$\boxed{u_\alpha = \frac{\Delta\alpha}{\Delta t}\cos\delta.}$$

Since both pieces are tiny angles, the curvature between them doesn't matter over the timescales we care about, so we can just add them with a plain old Pythagorean theorem instead of the full spherical law of cosines:

$$\boxed{u = \sqrt{\left(\frac{\Delta\delta}{\Delta t}\right)^2 + \left(\frac{\Delta\alpha}{\Delta t}\cos\delta\right)^2}.}$$

14.4 Equation of Time

Mean solar time (Section 12.3) pretends the Sun crawls across the sky at a perfectly steady 15° per hour. Reality is messier, for two reasons. One, Earth's orbit is an ellipse, so the Sun's angular speed along the ecliptic changes through the year. Two, the ecliptic is tilted against the celestial equator, so even a perfectly steady speed along the ecliptic gives a non steady rate of right ascension. Put those together and the Sun's real ("true") motion wobbles through the year, so **true solar time** and mean solar time slowly drift apart, by as much as 16 minutes. That drift is called the **equation of time**:

$$\boxed{E.T. = ST_{true} - ST_{mean}.}$$

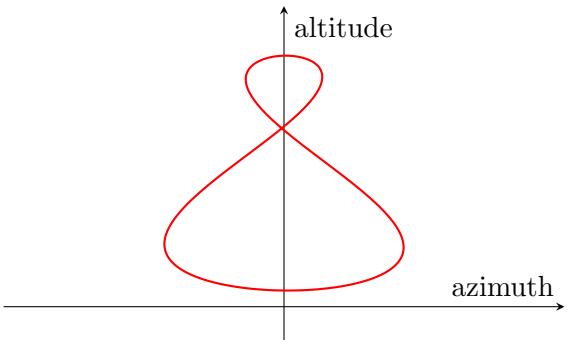


Figure 14: The analemma, the figure eight the Sun traces on the sky if you note its position at the same clock time every day for a year. It's a direct picture of the equation of time (its east-west width) mixed with the seasonal swing in declination (its north-south length).

15 Ecliptic and Galactic Coordinates: The Full Cast

Alright, you're a star now. Pick any one you like, say a dim one in Sagittarius. Four astronomers all wanna tell their friends where you are. The first one's standing on a hilltop and goes "about 30° up, a bit north of east." The second is flipping through a star atlas and goes "$\alpha = 17^h45^m$, $\delta = -29°$." The third hunts planets and goes "5° below the ecliptic, at longitude 270°." The fourth maps the Milky Way and goes "$l = 0°$, $b = +3°$." And the funny part is that all four are right. They just disagree on two things: what counts as *flat* and what counts as *zero*. This section is about those two choices, and how to translate between all four answers without losing a degree.

15.1 The Recipe for Any Coordinate System

Every spherical coordinate system on the sky is built from the same three decisions:

1. **Pick a fundamental plane.** It's a great circle that plays the role of the "equator" of the system.
2. **The pole comes for free.** It's the point 90° from every point on that plane. All you get to choose is which of the two poles is "north."
3. **Pick a zero point** on the fundamental plane, and a direction for angles to increase around the pole.

Then you measure two angles: a *latitude-like* one (how far above or below the plane you are, from $-90°$ to $+90°$) and a *longitude-like* one (how far around the pole you've gone from the zero point, $0°$ to $360°$, or 0^h to 24^h).

System	**Fundamental plane**	**Pole**	**Zero point**	**(lat-like, lon-like)**
Horizontal	the observer's horizon	zenith Z	north point of the horizon	altitude h, azimuth A (N→E)
Equatorial	celestial equator	NCP	vernal equinox γ	declination δ, right ascension α
Ecliptic	the ecliptic (Earth's orbital plane)	ecliptic pole K	vernal equinox γ	ecliptic latitude β, ecliptic longitude λ
Galactic	the mid-plane of the Milky Way	north galactic pole (NGP)	direction of the galactic center	galactic latitude b, galactic longitude l

Table 1: The four systems, each built from the same three choices. Notice that the equatorial and ecliptic systems share the same zero point, γ.

That last note is secretly the most useful thing in this section. Since the equatorial and ecliptic systems share the vernal equinox as their zero point, you can convert between them with just *one* rotation about the line through γ. The galactic system has nothing in common with the others, so it takes three rotations. We'll build exactly that in Section 15.5.

15.2 Horizontal and Equatorial: The Two Fundamental Systems, Recapped

You already met both of these in Section 3, so here's just the bit we need to compare them with the new guys.

Horizontal (h, A): your own personal sky. The fundamental plane is *your* horizon and the pole is *your* zenith. Great for pointing a telescope, useless for a catalogue, because it changes with where you are and with every minute of the night.

Equatorial (δ, α): the sky's own grid, hung on Earth's rotation axis. Declination δ is the sky's version of latitude, and right ascension α is its longitude, counted *eastward* from the vernal equinox. It stays put for a given star over human timescales, which is why catalogues use it. It's also the bridge to the observer, through the hour angle $H = LST - \alpha$ (Section 5).

The real difference between the four systems is *what each one is stuck to*:

System	Anchored to...
Horizontal	you, your location, and the time (it changes every minute)
Equatorial	Earth's spin axis (fixed against the stars, apart from slow precession)
Ecliptic	Earth's orbit around the Sun (the natural system for the Solar System)
Galactic	the Milky Way itself (the natural system for anything bigger than a few hundred parsecs)

15.3 Ecliptic Coordinates in Full

15.3.1 Definition

Now you're Earth, going around the Sun. From where you sit, the Sun creeps across the sky once a year along a great circle: the **ecliptic** (Section 9). Make that circle the fundamental plane and boom, you've got the **ecliptic system**:

- **Ecliptic latitude** β: the angle north (+) or south (−) of the ecliptic, $-90^\circ \le \beta \le +90^\circ$.
- **Ecliptic longitude** λ: the angle measured *eastward* along the ecliptic from the vernal equinox γ, $0^\circ \le \lambda < 360^\circ$.
- **Poles:** the north ecliptic pole K (in Draco) and the south ecliptic pole (in Dorado).

The Sun is *on* the ecliptic by definition, so $\beta_\odot = 0$ always, and $\lambda_\odot$ goes around from 0° to 360° at about $\dfrac{360^\circ}{365.2422\,\mathrm{d}} \approx 0.9856^\circ$ per day. (That's the formula $\lambda = \frac{n}{365.242} \cdot 360^\circ$ from Section 13.3.)

15.3.2 Where the Ecliptic Pole Sits

The ecliptic is tilted by the obliquity ϵ relative to the celestial equator, so its pole K is tilted by ϵ away from the NCP. But which way? The June solstice point $(\alpha, \delta) = (6^h, +\epsilon)$ is on the ecliptic, and K has to be 90° from it. Go straight over the pole from $(6^h, +\epsilon)$ and you land on the opposite hour circle, at polar distance ϵ from the NCP. So

$$\boxed{K: \quad (\alpha, \delta) = \Big(18^h,\ 90^\circ - \epsilon\Big) = \Big(18^h,\ +66.56^\circ\Big)\,.}$$

(Quick check: its polar distance from the NCP is ϵ, and its distance from the June solstice point is $(90^\circ - \epsilon) + \epsilon = 90^\circ$. ✓)

The four cardinal points of the ecliptic are worth memorizing in both languages:

$\lambda_\odot$	Event (northern hemisphere)	$\alpha_\odot$	$\delta_\odot$
0°	March (vernal) equinox	0^h	0°
90°	June solstice	6^h	$+\epsilon$
180°	September (autumnal) equinox	12^h	0°
270°	December solstice	18^h	$-\epsilon$

15.3.3 Why Solar System People Love It

Planets orbit in roughly the same plane as Earth, so from here they never wander far from the ecliptic. Their orbital inclinations to the ecliptic are tiny:

	Mercury	Venus	Mars	Jupiter	Saturn	Uranus	Neptune	Moon
i	7.0°	3.4°	1.85°	1.3°	2.5°	0.8°	1.8°	5.1°

So for a planet, β is just a small wobble, and λ tells you where it is along its path. That's also why the Moon keeps occulting the bright star Regulus, and why the zodiac constellations form a band. One subtlety: the **geocentric** ecliptic system is centered on Earth, while the **heliocentric** one is centered on the Sun. For stars it doesn't matter, but for planets it does. The two are linked by a simple relation: Earth's heliocentric longitude is $\lambda_\odot + 180°$ (seen from the Sun, Earth is on the opposite side of the sky from where the Sun appears when seen from Earth).

15.3.4 The Obliquity, Precession, and Why Catalogues Quote an Epoch

The obliquity isn't a constant. Today's value and its slow drift are

$$\boxed{\epsilon \approx 23.4393° - 0.0130°\, T,}$$

with T the number of Julian centuries since J2000.0 ($T = (JD - 2451545.0)/36525$), so ϵ is shrinking by about $47''$ per century. In this handout we mostly use 23.5° for quick estimates, but if you're quoting four digits, use 23.44°.

The bigger deal is **precession**. Earth's axis wobbles like a spinning top, tracing out a cone of half angle ϵ around K once every $\approx$ 25,772 years. The equator tilts along with it, so the vernal equinox slides *westward* along the ecliptic at about $50.3''$ per year. That has some consequences:

- λ of a fixed star goes up by $\approx 50.3''$ per year, but β doesn't change (the ecliptic plane itself barely moves).
- Both α and δ of a fixed star change, because the equator and the equinox both move.
- So catalogues state coordinates for an **epoch**, and the modern default is J2000.0.
- • You have to pick a new polar alignment star every few thousand years.

Why the pole star changes

The NCP circles K at an angular radius $\epsilon = 23.4°$. A star at angular distance d from K gets closest to the pole at $|d - \epsilon|$, so stars with d near ϵ are the real pole star candidates. Vega has $\beta \approx +61.7°$, so $d = 90° - 61.7° = 28.3°$, and the NCP will come within $28.3° - 23.4° \approx 4.8°$ of it in roughly 12,000 years. With ecliptic coordinates that's a two line problem.

15.4 Galactic Coordinates in Full

15.4.1 Definition

Okay, now step outside the Solar System. You're a star counter, and you care about where stuff sits relative to the *Milky Way*, because that's what organizes the sky: dust, young stars, globular clusters, and hydrogen gas all pile up near the disk. From inside the disk, it looks like a milky band running all the way around the sky, which is a great circle. Make that circle the fundamental plane.

- **Galactic latitude** b: the angle north (+) or south (−) of the galactic plane, $-90^\circ \le b \le +90^\circ$. Positive b points toward the north galactic pole (NGP), in Coma Berenices.
- **Galactic longitude** l: the angle around the NGP, measured from the direction of the **galactic center** (Sagittarius, $l = 0^\circ$), increasing in the same eastward sense as right ascension, $0^\circ \le l < 360^\circ$.

Handy landmarks: $l = 0^\circ$ is the galactic center. $l = 90^\circ$ is toward Cygnus, the direction the Sun is carried by galactic rotation. $l = 180^\circ$ is the anticenter (in Auriga). $l = 270^\circ$ is the direction the Sun came *from*, in the southern sky (near Vela).

15.4.2 The Defining Numbers

The galactic system is defined by three numbers that lock in its orientation relative to the equatorial system. In the modern (J2000) frame they are

$$\alpha_G = 12^h51^m26.28^s = 192.8595^\circ, \quad \delta_G = +27^\circ07'41.7'' = +27.1283^\circ \quad \text{(the NGP)},$$
$$l_{NCP} = 122.9319^\circ \quad \text{(the galactic longitude of the north celestial pole)}.$$

Everything else follows from these. For example:

- **Inclination:** the galactic plane is tilted $90^\circ - \delta_G = 62.87^\circ$ relative to the celestial equator (that's the angular distance from the NCP to the NGP).
- **Galactic center** ($l = b = 0$): $(\alpha, \delta) = (17^h45^m37.2^s,\ -28^\circ56'10'')$, in Sagittarius. The radio source Sgr A*, which marks the actual black hole, is at $(17^h45^m40.04^s, -29^\circ00'28.1'')$, about 0.07° away. That's close enough that the sample problem in Section 13.4 can safely treat it as "the center."
- **Nodes:** the galactic plane crosses the celestial equator at $\alpha = \alpha_G \pm 90^\circ$. The ascending node (where the plane crosses northward) is at $\alpha = 282.86^\circ = 18^h51^m$, which has galactic longitude $l_{NCP} - 90^\circ = 32.93^\circ$. The descending node is at $\alpha = 102.86^\circ = 6^h51^m$, $l = 212.93^\circ$.
- **The NCP in galactic coordinates:** $(l, b) = (122.93^\circ, +27.13^\circ)$. This is true by construction: the NCP's galactic latitude equals the NGP's declination.

15.4.3 The Three Planes Together

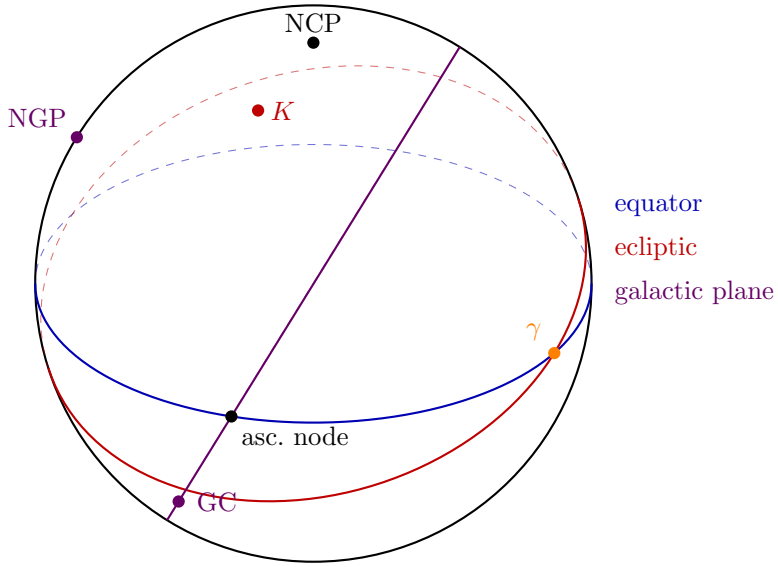


Figure 15: The three fundamental planes on the celestial sphere, viewed from a direction that lies in the galactic plane, so the Milky Way's disk (violet) is edge-on. Blue is the celestial equator. Red is the ecliptic, tilted 23.44° from the equator about the line through γ. Violet is the galactic plane, tilted 62.87° from the equator about the line of nodes (the ascending node is labelled). NCP, K, and NGP are the three poles, and GC marks the direction of the galactic center. Solid curves are the near side and dashed curves are the far side.

The angle between the ecliptic and the galactic plane is the angle between their poles, K and the NGP. We'll work it out in the sample problems below, and it comes out to about 60.2°. That tilt actually matters: the planets and the Moon wander along the ecliptic, which crosses the Milky Way at about 60° at two opposite points. One is near the direction of the galactic center (the Sun passes it every December), and the other is near the anticenter, on the Taurus and Gemini side of the sky.

Why bother with galactic latitude? The cosecant law

Dust in the disk dims background light. Model the dusty disk as a thin, plane parallel slab, and looking at galactic latitude b sends your line of sight through a path length $\propto 1/\sin|b|$ of dust, so the extinction scales as

$$A(b) \approx \frac{A_\perp}{\sin|b|},$$

where $A_\perp$ is the extinction straight out of the disk. In galactic coordinates that's a one variable problem. In equatorial coordinates the disk is a tilted, wavy curve and the same statement is a mess. (The slab model falls apart really close to the plane, for $|b| \lesssim 10°$, where the geometry isn't a slab anymore.)

15.5 Transformations: The Rotation Method

Sections 12 to 14 got all our formulas out of spherical triangles, one at a time. Here's the bird's eye view: *every* transformation between these systems is just a rotation of 3D axes.

Start by packing a direction on the sky into a unit vector, using the longitude ℓ and latitude ϕ' of whichever system you're in:

$$\hat{r} = \begin{pmatrix} \cos\phi' \cos\ell \\ \cos\phi' \sin\ell \\ \sin\phi' \end{pmatrix}.$$

So in equatorial coordinates $\hat{r}_{eq} = (\cos\delta\cos\alpha,\ \cos\delta\sin\alpha,\ \sin\delta)$, in ecliptic coordinates $\hat{r}_{ecl} = (\cos\beta\cos\lambda,\ \cos\beta\sin\lambda,\ \sin\beta)$, and in galactic coordinates $\hat{r}_{gal} = (\cos b\cos l,\ \cos b\sin l,\ \sin b)$. Switching systems is just rotating the axes. We'll use two basic "axis rotation" matrices (they give the coordinates of a fixed point as the axes turn by θ):

$$R_x(\theta) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & \sin\theta \\ 0 & -\sin\theta & \cos\theta \end{pmatrix}, \qquad R_z(\theta) = \begin{pmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Rotations are orthogonal matrices, so the inverse of a rotation is just its transpose, which is the same as rotating by $-\theta$.

15.5.1 Equatorial ↔ Ecliptic: One Rotation

Both systems share the x axis (the vernal equinox), and the ecliptic pole is tilted by ϵ from the equatorial pole, so one rotation about x does it:

$$\boxed{\hat{r}_{ecl} = R_x(\epsilon)\,\hat{r}_{eq}, \qquad \hat{r}_{eq} = R_x(-\epsilon)\,\hat{r}_{ecl}.}$$

Multiply it out row by row and you get

$$\begin{aligned} \cos\beta\cos\lambda &= \cos\delta\cos\alpha, \\ \cos\beta\sin\lambda &= \sin\delta\sin\epsilon + \cos\delta\cos\epsilon\sin\alpha, \\ \sin\beta &= \sin\delta\cos\epsilon - \cos\delta\sin\epsilon\sin\alpha. \end{aligned}$$

Those are exactly the formulas from Sections 13.1 and 14.2, but now you can see them as three parts of one operation instead of a bunch of separate triangle problems. The inverse rotation gives $\sin\delta = \cos\epsilon\sin\beta + \sin\epsilon\cos\beta\sin\lambda$ from Section 13.2, plus the two companions $\cos\delta\cos\alpha = \cos\beta\cos\lambda$ and $\cos\delta\sin\alpha = \cos\epsilon\cos\beta\sin\lambda - \sin\epsilon\sin\beta$.

Always use `atan2`, not $\tan^{-1}$ of a ratio

To get angles back out of a vector, use latitude $= \sin^{-1}(z)$ and longitude $= \mathrm{atan2}(y, x)$. If all you have is the ratio y/x, a plain $\tan^{-1}$ can't tell the second quadrant from the fourth, and you're off by 180°. The same trap is hiding in $\tan\alpha_\odot = \cos\epsilon\tan\lambda_\odot$ from Section 13.3. It has two solutions 180° apart, and the right one is always in the *same quadrant* as $\lambda_\odot$. Check the signs of sin and cos separately before you trust your calculator.

15.5.2 Equatorial ↔ Galactic: Three Rotations

The galactic frame is three turns away from the equatorial one. Here's how you'd physically get there, starting from the equatorial frame:

1. Spin about the z axis by $\alpha_G + 90°$, so the x axis points at the galactic plane's ascending node.
2. Tilt about the new x axis by $90° - \delta_G$, so the equatorial plane lays down onto the galactic plane.
3. Spin about the new z axis (now the galactic pole) by $90° - l_{NCP}$, so the x axis points at the galactic center instead of the node. (The node sits at $l = l_{NCP} - 90°$, so we have to undo that offset.)

In matrix form,

$$\boxed{\hat{r}_{gal} = R_z(90° - l_{NCP})\; R_x(90° - \delta_G)\; R_z(\alpha_G + 90°)\; \hat{r}_{eq}.}$$

Plug in the J2000 numbers and you get one fixed matrix $\mathbf{T}$:

$$\hat{r}_{gal} = \mathbf{T}\,\hat{r}_{eq}, \qquad \mathbf{T} = \begin{pmatrix} -0.0549 & -0.8734 & -0.4838 \\ +0.4941 & -0.4448 & +0.7470 \\ -0.8677 & -0.1981 & +0.4560 \end{pmatrix}, \qquad \hat{r}_{eq} = \mathbf{T}^{\mathsf{T}}\hat{r}_{gal}.$$

The *rows* of $\mathbf{T}$ mean something nice: they're the new axes written in the old coordinates. The third row, $(-0.8677, -0.1981, 0.4560)$, is exactly $(\cos\delta_G\cos\alpha_G,\ \cos\delta_G\sin\alpha_G,\ \sin\delta_G)$, the unit vector toward the NGP. Makes sense, right? $\sin b$ is just how far you are along the galactic pole. The first row is the equatorial unit vector toward the galactic center.

15.5.3 The Spherical Triangle Version (for Hand Calculation)

If matrices aren't your thing, the same transformation comes out of the triangle made by the NCP, the NGP, and your object, where the side between the NCP and the NGP is $90° - \delta_G$:

$$\boxed{\begin{aligned} \sin b &= \sin\delta\sin\delta_G + \cos\delta\cos\delta_G\cos(\alpha - \alpha_G), \\ \cos b\,\sin(l_{NCP} - l) &= \cos\delta\sin(\alpha - \alpha_G), \\ \cos b\,\cos(l_{NCP} - l) &= \sin\delta\cos\delta_G - \cos\delta\sin\delta_G\cos(\alpha - \alpha_G). \end{aligned}}$$

Solve the first one for b, then use the last two together (with atan2) to get l. Going the other way,

$$\boxed{\begin{aligned} \sin\delta &= \sin b\sin\delta_G + \cos b\cos\delta_G\cos(l_{NCP} - l), \\ \cos\delta\,\sin(\alpha - \alpha_G) &= \cos b\sin(l_{NCP} - l), \\ \cos\delta\,\cos(\alpha - \alpha_G) &= \sin b\cos\delta_G - \cos b\sin\delta_G\cos(l_{NCP} - l). \end{aligned}}$$

15.6 Cheat Sheet

Quantity	Value (J2000)
Obliquity of the ecliptic	$\epsilon = 23.4393^\circ$
North ecliptic pole K	$(\alpha, \delta) = (18^h, +66.56^\circ)$
North galactic pole	$(\alpha_G, \delta_G) = (12^h51^m26.28^s, +27.1283^\circ)$
l of the north celestial pole	$l_{NCP} = 122.9319^\circ$
Galactic center ($l = b = 0$)	$(17^h45^m37.2^s, -28.9362^\circ)$
Angle between equator and galactic plane	62.87°
Angle between ecliptic and galactic plane	60.19°
Galactic plane ascending node	$\alpha = 18^h51^m$, $l = 32.93^\circ$
K in galactic coordinates	$(l, b) = (96.38^\circ, +29.81^\circ)$

15.7 Sample Problems

Problem: Regulus in Ecliptic and Galactic Coordinates

Regulus has $(\alpha, \delta) = (10^h08^m22.3^s, +11^\circ58'02'')$. Find its ecliptic and galactic coordinates.

Convert to decimal degrees: $\alpha = 152.093^\circ$, $\delta = +11.967^\circ$. For the ecliptic, use the $R_x(\epsilon)$ rows with $\epsilon = 23.4393^\circ$. Latitude first: $\sin\beta = \cos\epsilon\sin\delta - \sin\epsilon\cos\delta\sin\alpha \approx 0.1902 - 0.1822 = 0.0080$, so $\beta \approx +0.46^\circ$. For the longitude, the two companion rows give $\cos\beta\cos\lambda = \cos\delta\cos\alpha \approx -0.8645$ and $\cos\beta\sin\lambda = \sin\delta\sin\epsilon + \cos\delta\cos\epsilon\sin\alpha \approx +0.5027$. Cosine is negative and sine is positive, so λ is in the second quadrant: $\lambda = \text{atan2}(0.5027, -0.8645) \approx 149.8^\circ$. (A bare $\tan^{-1}$ would've given you -30.2°, which is off by exactly 180°. Classic.) So $(\lambda, \beta) \approx (149.8^\circ, +0.46^\circ)$. Regulus sits less than half a degree from the ecliptic, which is exactly why planets and the Moon keep passing so close to it. Run the matrix $\mathbf{T}$ and you get $(l, b) \approx (226.4^\circ, +48.9^\circ)$. That's way above the galactic plane, which is why we can see so many galaxies over there (not much dust in the way).

Problem: The Angle Between the Ecliptic and the Galactic Plane

Find the angle between the plane of the ecliptic and the plane of the Milky Way, and the galactic coordinates of the north ecliptic pole K.

The angle between two planes is the angle between their poles. The pole K has equatorial coordinates $(18^h, +66.56^\circ)$, so $\alpha - \alpha_G = 270^\circ - 192.86^\circ = 77.14^\circ$. The first transformation equation gives

$$\sin b_K = \sin 66.56^\circ \sin 27.13^\circ + \cos 66.56^\circ \cos 27.13^\circ \cos 77.14^\circ \approx 0.497,$$

so $b_K \approx +29.8^\circ$. Now, b_K is how far K sits above the galactic plane, so its distance from the galactic pole is $90^\circ - b_K$, and that distance is the angle between the planes:

$$\theta_{ecl,gal} = 90^\circ - 29.8^\circ \approx \boxed{60.2^\circ}.$$

Feed the other two equations into atan2 and you get $l_K \approx 96.4^\circ$. By the same symmetry, the NGP sits at ecliptic latitude $+29.8^\circ$.

Problem: The Galactic Center from Quezon City

What is the maximum altitude of the galactic center for an observer at $\phi = 14.5°N$, from what latitude would it pass through the zenith, and beyond which northern latitude is it never visible?

The galactic center has $\delta_{GC} = -28.94°$. At upper culmination (Section 6.2), $h_u = 90° - |\phi - \delta| = 90° - |14.5° + 28.94°| = 46.6°$. It goes through the zenith when $\phi = \delta_{GC} = 28.94°S$. It never rises when $h_u < 0$, meaning $|\phi - \delta| > 90°$, which gives $\phi > 90° - 28.94° = 61.06°N$. (That's why the galactic center is such a southern hemisphere flex.)

16 How Long Is a Day? How Long Is a Year?

Okay, now you're the Earth. You're doing two things at once: spinning on your axis and falling around the Sun. Neither one of those has ever heard of a clock. So when somebody asks "how long is a day?" the honest answer is "*compared to what?*" Compared to the faraway stars? The Sun? The equinox? Every one of them gives you a different number. This section is about getting all of them right, and about the tilt of your axis, ϵ, which turns out to run the whole show. It decides how long daylight lasts, why the solar day isn't 24 hours, and even where the climate zones end.

16.1 A Zoo of Days and Years

Name	**Defined as the time for...**	**Length**
Sidereal day	Earth to rotate 360° relative to the equinox (stars)	$23^h56^m4.09^s = 86164.09$ s
Mean solar day	Sun to return to the meridian, averaged over the year	$24^h = 86400$ s
Apparent solar day	Sun to return to the meridian, on a specific date	24^h from -21 s to $+30$ s
Sidereal year	Earth to return to the same place among the stars	365.25636 d
Tropical year	Sun to return to the vernal equinox (the year of the seasons)	365.24219 d
Anomalistic year	Earth to return to perihelion	365.25964 d

Table 2: Days and years. (The Julian calendar uses 365.25 *d and the Gregorian* 365.2425 *d. See the aside at the end of Section 16.)*

The anomalistic year comes out a few minutes longer than the sidereal one because the orbit's perihelion itself creeps eastward by roughly $11.6''$ per year, so Earth has to go a tiny bit past one full lap to get back to it.

16.2 Sidereal Day and Solar Day

So why is a solar day longer than a sidereal day? Because while you're spinning, you're also moving around the Sun. After one full 360° spin relative to the stars, the Sun looks like it has shifted eastward by about 1° (that's just your own orbital motion), so you have to spin an extra degree before the Sun's back on the meridian.

Let's make that exact. Let T_{rot} be the rotation period relative to the stars and P the orbital period, with the spin and the orbit going the same direction (prograde). In a time t you turn through $\omega_{rot}t$ and move around the orbit through Ωt, where $\omega_{rot} = 2\pi/T_{rot}$ and $\Omega = 2\pi/P$. The Sun is back on the meridian once the spin has beaten the orbit by exactly one full turn:

$$\omega_{rot}t - \Omega t = 2\pi \quad \Longrightarrow \quad \boxed{\frac{1}{T_{sol}} = \frac{1}{T_{rot}} - \frac{1}{P} \qquad \text{(prograde spin).}}$$

If the planet spins backwards (retrograde), the two angular speeds add up instead:

$$\boxed{\frac{1}{T_{sol}} = \frac{1}{T_{rot}} + \frac{1}{P} \qquad \text{(retrograde spin).}}$$

Multiply through by P and you get a counting rule that's super easy to remember:

$$\frac{P}{T_{sol}} = \frac{P}{T_{rot}} \mp 1.$$

The number of solar days in a year is one less than the number of sidereal days (prograde) or one more (retrograde). For Earth, a year has 365.256 solar days and 366.256 sidereal days, so

$$T_{sid} = \frac{365.25636}{366.25636} \times 86400 \text{ s} = 86164.10 \text{ s},$$

which is $23^h56^m4.10^s$. (Small footnote: if you measure the sidereal day relative to the *equinox* instead of the stars, you have to use the tropical year, and you get $\frac{365.24219}{366.24219} \times 86400 = 86164.09$ s. That 0.008 s difference comes from precession, and it's the only reason the table above says 4.09 s.)

16.2.1 Other Worlds

	T_{rot}	Spin	P	T_{sol}
Mercury	58.646 d	prograde	87.969 d	175.94 d
Venus	243.0185 d	retrograde	224.701 d	116.75 d
Mars	24.6229 h	prograde	686.98 d	24.660 h = $24^h39^m35^s$

Mercury's solar day is exactly two of its years long, which is kinda wild. Its 3:2 spin orbit resonance makes $P = 1.5\,T_{rot}$, so $T_{sol} = 3T_{rot} = 2P$. On Venus, the retrograde spin is why the Sun rises in the west, and it's also why the solar day is *shorter* than the sidereal one.

16.3 Sidereal Year and Tropical Year: Precession at Work

The Sun's ecliptic longitude is measured from the vernal equinox, and the equinox slides westward along the ecliptic at $p = 360°/T_{prec}$ with $T_{prec} \approx 25{,}772$ yr. So the Sun, creeping eastward, bumps into the equinox a little *early*: it only has to cover $360° - p\,t$ instead of $360°$. Write $\omega = 2\pi/T_{sid}$ for Earth's orbital angular speed, and then

$$\omega\,t = 2\pi - \frac{2\pi}{T_{prec}}t \text{ (the equinox slides forward to meet the Sun)} \Longrightarrow \boxed{\frac{1}{T_{trop}} = \frac{1}{T_{sid}} + \frac{1}{T_{prec}}.}$$

Plug in $T_{sid} = 365.25636$ d and $T_{prec} = 25{,}772 \times 365.2422$ d:

$$T_{trop} = \frac{T_{sid}}{1 + T_{sid}/T_{prec}} \approx 365.25636 - 0.01417 = 365.24219 \text{ d}.$$

So the tropical year is shorter than the sidereal year by 0.01417 d $\approx 20^m 24^s$. That's the number that matters for a calendar, because seasons follow the equinox, not the stars.

So where does obliquity come in? Precession is a torque thing. The Sun and Moon tug on Earth's equatorial bulge, which is tilted by ϵ from the orbital plane, and they're basically trying to straighten the spin axis up. The torque is proportional to $\sin\epsilon\cos\epsilon$, and the axis reacts like a gyroscope: divide by the spin angular momentum times $\sin\epsilon$ and you're left with a precession rate proportional to $\cos\epsilon$, around a cone of half angle ϵ. If ϵ were zero, the cone would shrink to nothing and the axis would just stay put.

16.4 Day Length and Obliquity

How long is the Sun actually above the horizon? We already have the key equation from Section 12.1.3,

$$\cos H = -\tan\phi\tan\delta,$$

where H is the Sun's hour angle at sunrise or sunset. The Sun spends a total of $2H$ of hour angle above the horizon, and 15° of hour angle is 1 hour, so

$$\boxed{D = \frac{2H}{15^\circ/\text{hr}} = \frac{H}{7.5^\circ}\ \text{hours.}}$$

The obliquity sets $\delta_\odot$ through the formula from Section 13.3, $\sin\delta_\odot = \sin\epsilon\sin\lambda_\odot$, with $\lambda_\odot = 360^\circ n/365.242$ and n the number of days since the March equinox. Put it all together and the daylight at any latitude on any date comes out as one closed formula:

$$\boxed{D(\phi,n) = \frac{1}{7.5^\circ}\cos^{-1}\Big[-\tan\phi\,\tan\Big(\sin^{-1}\Big(\sin\epsilon\,\sin\tfrac{360^\circ n}{365.242}\Big)\Big)\Big]\ \text{hours.}}$$

(If the thing in the square brackets is above +1, the Sun never rises: polar night, $D = 0$. If it's below −1, the Sun never sets: polar day, $D = 24$ h.)

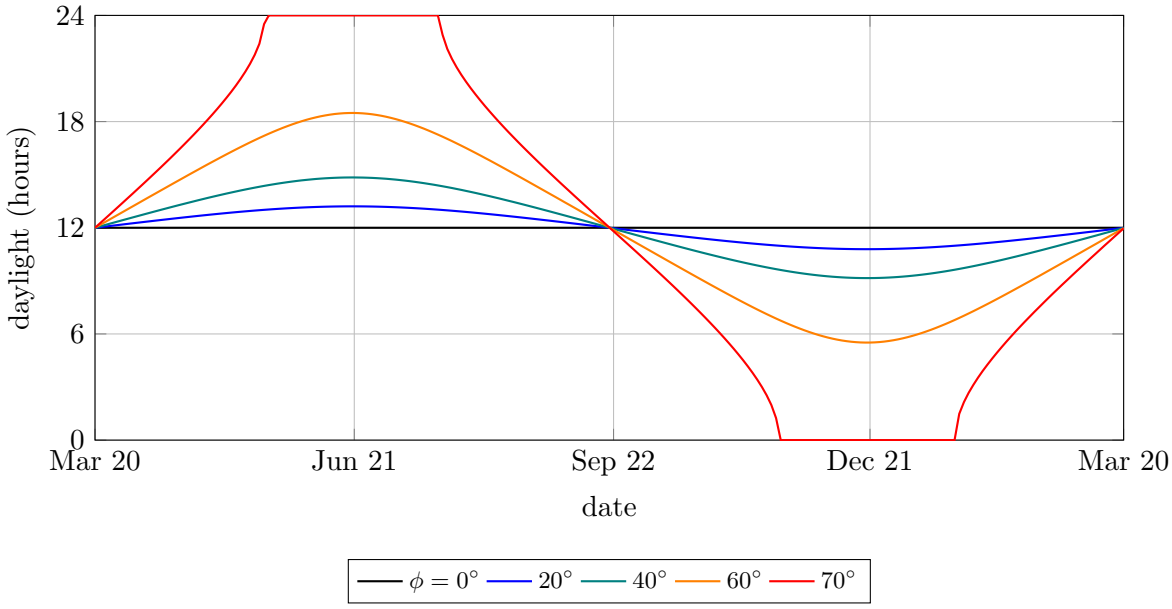


Figure 16: Hours of daylight through the year at several northern latitudes ($\epsilon = 23.44°$). The equator is stuck at 12 h, every curve passes through 12 h at the two equinoxes, and the higher the latitude, the bigger the swing. At 70° the flat top is polar day and the flat bottom is polar night.

A few things fall right out of this:

- **On the equator** ($\phi = 0$): $\cos H = 0$, so $H = 90°$ and $D = 12$ h *every single day*, for any obliquity.
- **At an equinox** ($\delta = 0$): $\cos H = 0$ again, so $D = 12$ h at every latitude. (That's literally what the word "equinox" means: "equal night.")
- **At a solstice** ($\delta = \pm\epsilon$): $\cos H_{max} = -\tan\phi \tan\epsilon$. The longest and shortest days are mirror images: $D_{short} = 24\text{ h} - D_{long}$.
- **Polar day or night** happens when $|\tan\phi \tan\delta| > 1$, which means $|\delta| > 90° - |\phi|$. Since δ tops out at ϵ, the Sun can only refuse to set poleward of $\boxed{\phi_{polar} = 90° - \epsilon = 66.56°}$. Those are the Arctic and Antarctic Circles.
- **The tropics** are the flip side: the Sun can only be straight overhead at latitudes $|\phi| \le \epsilon$, which is why the tropics of Cancer and Capricorn sit at $\pm 23.44°$.

So the obliquity literally *draws* the climate zone lines on a planet.

Other planets: when the tilt is more than 90°

The formula $\sin\delta_\odot = \sin\epsilon \sin\lambda_\odot$ works for any obliquity, so $\delta_{max} = \sin^{-1}(\sin\epsilon)$. For $\epsilon \le 90°$ that's just ϵ, and for $\epsilon > 90°$ it's $180° - \epsilon$. Define the *effective* tilt $\epsilon' = \min(\epsilon, 180° - \epsilon)$. Then the tropics sit at $\pm\epsilon'$ and the polar circles sit at $\pm(90° - \epsilon')$.

	ϵ	ϵ'	tropics at	polar circles at
Mercury	0.03°	0.03°	0.03°	89.97°
Venus	177.4°	2.6°	2.6°	87.4°
Earth	23.44°	23.44°	23.44°	66.56°
Mars	25.19°	25.19°	25.19°	64.81°
Jupiter	3.13°	3.13°	3.13°	86.87°
Saturn	26.73°	26.73°	26.73°	63.27°
Uranus	97.77°	82.23°	82.23°	7.77°
Neptune	28.32°	28.32°	28.32°	61.68°

Uranus is the extreme case. Its tropics reach to within 8° of the poles, and its "arctic circles" are only 7.8° from the equator, so a huge chunk of the planet gets polar day and polar night at some point during its 84 year orbit. Absolute chaos.

16.4.1 A More Honest Sunrise

The formula above treats the Sun like a single point and ignores the atmosphere. In real life, sunrise is when the *top edge* of the Sun touches the horizon. Refraction lifts the Sun by about $34'$ near the horizon (Section 12.4), and the Sun's apparent radius is about $16'$, so the Sun's *center* is at altitude $h_0 \approx -(34' + 16') = -50' = -0.833°$ at the moment of sunrise. Swap $h = 0$ for $h = h_0$ in the altitude formula from Section 12.1.1 and you get

$$\boxed{\cos H = \frac{\sin h_0 - \sin\phi \sin\delta}{\cos\phi \cos\delta}, \qquad h_0 = -0.833°.}$$

This stretches the day a little. At the equator on an equinox, $\cos H = \sin(-0.833°) = -0.0145$, so $H = 90.83°$, and $D = 12.11$ h $= 12^h07^m$ instead of exactly 12 h.

16.5 Why the Solar Day Is Not Exactly 24 Hours

Okay, this is where obliquity really gets its hands dirty. Take two successive meridian passages of the Sun. Between them, Earth has turned through 360° (relative to the equinox) *plus* however far the Sun's right ascension has crept eastward in the meantime. Call that extra bit $\Delta\alpha$. Then

$$T_{true} = T_{sid}\left(1 + \frac{\Delta\alpha}{360°}\right).$$

If the Sun moved at a steady pace along the *equator*, you'd get $\Delta\alpha = 360°/365.2422$ every day, and $T = 86164.09 \times (1 + 1/365.2422) = 86400.00$ s. That's where the 24 hours comes from. But the Sun doesn't move along the ecliptic, which is tilted by ϵ, and what matters for the meridian is the *right ascension* it picks up each day.

The obliquity effect. Say the Sun moves at a steady pace along the ecliptic, so λ grows at a constant rate. Differentiate $\tan\alpha = \cos\epsilon \tan\lambda$:

$$\sec^2\alpha\, d\alpha = \cos\epsilon \sec^2\lambda\, d\lambda \implies \frac{d\alpha}{d\lambda} = \frac{\cos\epsilon}{\cos^2\lambda + \cos^2\epsilon \sin^2\lambda} = \frac{\cos\epsilon}{1 - \sin^2\epsilon \sin^2\lambda} = \boxed{\frac{\cos\epsilon}{\cos^2\delta_\odot}}.$$

(The last step uses $\sin\delta_\odot = \sin\epsilon \sin\lambda$.) Two special cases tell you the whole story:

- **Equinoxes** ($\delta = 0$): $d\alpha/d\lambda = \cos\epsilon < 1$. The ecliptic is tilted against the equator there, so moving 1° along it only gets you $\cos\epsilon°$ of right ascension. The Sun's R.A. crawls, so the solar day is *shorter*.
- **Solstices** ($\delta = \pm\epsilon$): $d\alpha/d\lambda = 1/\cos\epsilon > 1$. There the ecliptic runs parallel to the equator, and the lines of constant R.A. are squished together near the pole, so the Sun's R.A. races, and the solar day is *longer*.

Throw in numbers, using $T_{sid}/365.2422 = 235.91$ s:

$$\boxed{\Delta T_{obl} = 235.91 \text{ s} \times \left(\frac{\cos\epsilon}{\cos^2\delta_\odot} - 1 \right).}$$

At the equinoxes, $235.91 \times (\cos 23.44° - 1) = -19.5$ s. At the solstices, $235.91 \times (1/\cos 23.44° - 1) = +21.2$ s.

The eccentricity effect. Earth's orbit is an ellipse with $e = 0.0167$, and by Kepler's second law Earth moves faster near perihelion (early January). To first order in e, $\dot\lambda = \dot\lambda_{mean}(1 + 2e\cos M)$, where M is the mean anomaly ($M = 0$ at perihelion). That scales the daily R.A. gain by $(1 + 2e\cos M)$, so

$$\Delta T_{ecc} \approx 235.91 \text{ s} \times 2e\cos M,$$

which is +7.9 s at perihelion and −7.9 s at aphelion.

Now smash them together (multiply the two corrections and keep first order terms):

$$\boxed{\Delta T \approx 235.91 \text{ s} \times \left[\frac{\cos\epsilon}{\cos^2\delta_\odot}\,(1 + 2e\cos M) - 1 \right].}$$

Let's see if it matches reality. Around December 22, perihelion is only about 12 days away ($M \approx -12°$, so $1 + 2e\cos M \approx 1.0327$) and the Sun is near its solstice ($\cos\epsilon/\cos^2\delta_\odot \approx 1.0894$), so $\Delta T \approx 235.91 \times [1.0894 \times 1.0327 - 1] \approx +29.5$ s. The longest solar day of the year is about $24^h0^m30^s$, exactly as observed. Around September 17, $M \approx 254°$ (so $1 + 2e\cos M \approx 0.9905$) and $\delta_\odot \approx +0.6°$ ($\cos\epsilon/\cos^2\delta_\odot \approx 0.9176$), so $\Delta T \approx 235.91 \times [0.9176 \times 0.9905 - 1] \approx -21.5$ s. The shortest solar day is about $23^h59^m39^s$, also as observed. Both predictions land within about a

second of the real values, which is honestly kinda satisfying.

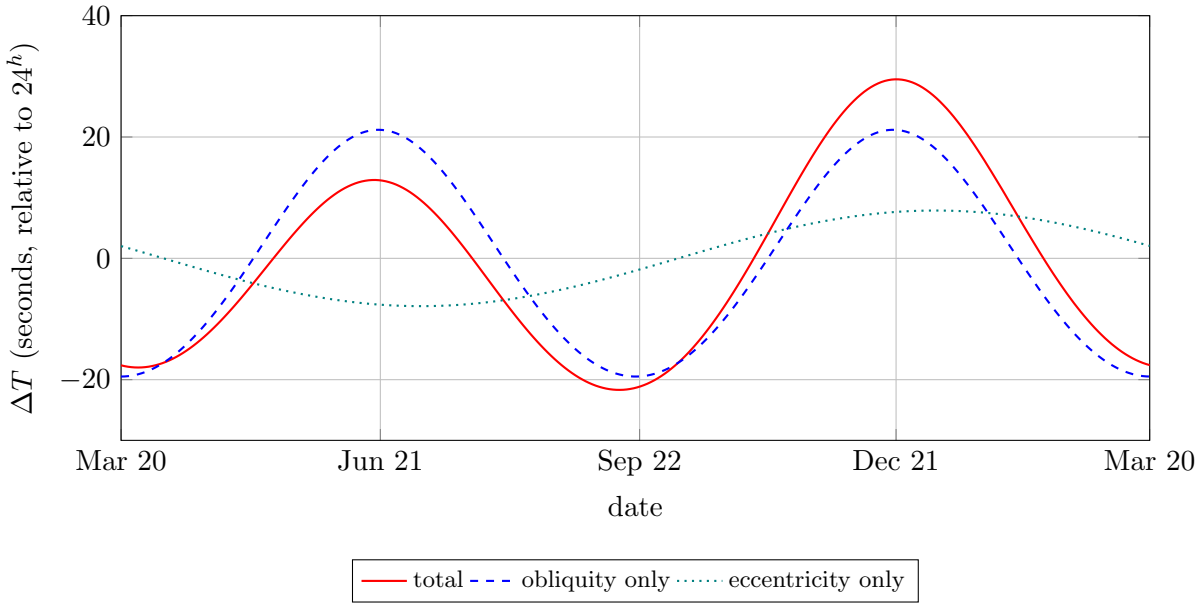


Figure 17: How much longer (+) or shorter (−) than 24^h the true solar day is, through the year, from the first order formula above. The obliquity term (dashed) oscillates twice a year, and the eccentricity term (dotted) once a year. Their sum (solid) peaks near +30 s around December 20 and bottoms out near −22 s around mid September.

How this connects to the equation of time. A sundial advances 24^h per *true* solar day. If a true day lasts ΔT seconds longer than 24 clock hours, the sundial loses ΔT seconds against the clock that day. With the convention $E.T. = ST_{true} - ST_{mean}$ from Section 14.4,

$$\frac{d(E.T.)}{dn} = -\Delta T \text{ (seconds per day)}.$$

So the equation of time is just the running total of Figure 17, flipped upside down. That also explains something that trips up a lot of people: $E.T.$ hits its extremes when ΔT crosses *zero* (the sundial stops drifting), not at the solstices where ΔT peaks. So the $E.T.$ extremes of about +16 min in early November and −14 min in mid February, plus the smaller wiggles in May and July, are exactly where Figure 17's curve crosses zero.

Calendars, and why the seasons aren't equal

Calendars. The seasons follow the tropical year, 365.24219 d. The Julian calendar's 365.25 d overshoots by 0.0078 d/yr, which is a one day error every ≈ 128 years, and it had piled up about 10 days by 1582. The Gregorian rule (a leap year every 4 years, except century years not divisible by 400) has 97 leap years per 400, so its mean year is $365 + 97/400 = 365.2425$ d. That's off by only 0.0003 d/yr, or one day in about 3,200 years.

Seasons. Since Earth moves fastest near perihelion (early January), northern winter is the shortest season. The astronomical seasons in the northern hemisphere last roughly spring 92.8 d, summer 93.6 d, autumn 89.8 d, and winter 89.0 d, which adds up to 365.2 d. Same orbit, same obliquity, different lengths. That's Kepler's second law doing its thing.

16.6 Sample Problems

Problem: Solar Day on Mars

Mars rotates once relative to the stars in 24.6229 hours and orbits the Sun in 686.98 days. What is the length of a Martian solar day (a sol)?

Mars spins prograde, so $1/T_{sol} = 1/T_{rot} - 1/P$. Convert the orbital period to hours: $P = 686.98 \times 24 = 16487.5$ h. Then

$$\frac{1}{T_{sol}} = \frac{1}{24.6229} - \frac{1}{16487.5} = 0.040613 - 0.0000607 = 0.040553 \text{ h}^{-1},$$

so $T_{sol} = 24.660$ h $= 24^h 39^m 35^s$. A sol is about 2.7% longer than an Earth day.

Problem: The Solar Day on Venus

Venus spins retrograde with $T_{rot} = 243.0185$ d and orbits in 224.701 d. How long does the Sun take to go from one noon to the next, and how many such days are in a Venusian year?

It's retrograde, so $1/T_{sol} = 1/243.0185 + 1/224.701 = 0.0041152 + 0.0044504 = 0.0085656$ d^{-1}, which gives $T_{sol} = 116.75$ d. The number of solar days per year is $P/T_{sol} = 224.701/116.75 = 1.92$, which matches the counting rule $P/T_{rot} + 1 = 0.925 + 1 = 1.925$.

Problem: Daylight at Manchester

Find the length of daylight at $\phi = 53.48°N$ at the June and December solstices (ignore refraction and the size of the Sun), and the latitude at which the longest day lasts exactly 18 hours.

At the June solstice $\delta = +\epsilon = 23.44°$, so $\cos H = -\tan 53.48° \tan 23.44° = -1.3505 \times 0.4336 = -0.5856$. That gives $H = 125.8°$ and $D = H/7.5° = 16.78$ h $\approx 16^h 47^m$. At the December solstice $\delta = -\epsilon$, so the sign of $\cos H$ flips, $H = 54.2°$, and $D = 24 - 16.78 = 7.22$ h $\approx 7^h 13^m$. (If you include refraction and the Sun's size with $h_0 = -0.833°$, the June day gets about a quarter of an hour longer.)

For an 18 h day, $H = 135°$ and $\cos H = -0.7071$, so $\tan\phi \tan\epsilon = 0.7071$. That gives $\tan\phi = 0.7071/0.4336 = 1.631$ and $\phi = 58.5°$ N.

Problem: A Mars-Like Planet

A planet has obliquity $\epsilon = 25.19°$. What is the longest daylight at $\phi = 45°$ (in hours of 1/24 of its solar day), and at what latitudes does the Sun go all the way to the zenith at least once?

At the solstice, $\cos H = -\tan 45° \tan 25.19° = -0.4706$, so $H = 118.1°$ and $D = H/7.5° = 15.74$ "hours." The Sun only gets straight overhead when $|\phi| \le \epsilon'$, which means anywhere between 25.19° S and 25.19° N.

17 Time, Properly: Julian Dates, Zone Time, and Sidereal Time

Okay, real talk. In an actual contest problem, nobody hands you an hour angle. They hand you a date, a clock reading, and a city, and you have to turn that into something you can plug into $\sin h = \sin\delta\sin\phi + \cos\delta\cos\phi\cos H$. That translation step is where a ridiculous number of points get lost, so this whole section is about doing it cleanly.

17.1 Julian Date

Calendars are a mess (months of different lengths, leap years, you name it), so astronomers just count days. The **Julian Date (JD)** is the number of days since noon UT on January 1, 4713 BC (in the old Julian calendar). Yes, really. It makes subtracting two dates trivial, which is the whole point. Two things to remember: a JD day *starts at noon*, so midnight UT always ends in .5, and the modern reference epoch is **J2000.0**, which is $JD = 2451545.0$ (2000 January 1, 12:00).

To get the JD from a Gregorian calendar date (year Y, month M, day D, where D includes the fraction of the day, like 10.80625 for 19:21 UT on the 10th): if $M \le 2$, replace $Y \to Y-1$ and $M \to M+12$. Then

$$A = \lfloor Y/100 \rfloor, \qquad B = 2 - A + \lfloor A/4 \rfloor,$$

$$\boxed{JD = \lfloor 365.25\,(Y+4716) \rfloor + \lfloor 30.6001\,(M+1) \rfloor + D + B - 1524.5.}$$

(The floor brackets $\lfloor\ \rfloor$ just mean "round down." This works for any Gregorian date after 1582 October 15.) Two handy cousins: the **Julian century** $T = (JD - 2451545.0)/36525$, which is the T in the obliquity formula from Section 15, and the **Modified Julian Date** $MJD = JD - 2400000.5$, which just has smaller numbers and starts at midnight.

A quick word on UT, UTC, and TT

Your clock runs on **UTC**. Earth's actual rotation angle is **UT1**, and leap seconds keep the two within 0.9 s of each other, so for sidereal time problems you can treat UTC as UT. Ephemerides (the tables that say where the planets are) use **Terrestrial Time (TT)**, which is a perfectly uniform time scale and currently runs ahead of UTC by about 69 s. For anything in this handout, that difference doesn't matter.

17.2 Local Mean Time, Zone Time, and the Equation of Time

Remember mean solar time from Section 12.3? It's tied to *your longitude*. The Earth turns 15° per hour, so every 15° east means your local mean time is one hour ahead of UT:

$$\boxed{LMT = UT + \frac{\lambda}{15^\circ}\ \text{hours}, \qquad (\lambda \text{ positive east}).}$$

Nobody lives by that, because every town would have its own time. Instead we have time zones, where the clock reads $UT+$ a whole number of hours (usually). If your zone's standard meridian is at λ_{std}, the difference between your local mean time and your clock is

$$LMT - (\text{zone time}) = \frac{\lambda - \lambda_{std}}{15^\circ}\ \text{hours} = 4\ \text{minutes per degree}.$$

Then there's the Sun itself. A sundial reads *apparent* solar time, which differs from mean time by the equation of time from Section 14.4:

$$\text{apparent time} = \text{mean time} + E.T.$$

So the Sun crosses your meridian (apparent noon) when *mean* time reads $12^h - E.T.$

A closed form for E.T. Section 16.5 split the equation of time into an eccentricity piece and an obliquity piece. Let's turn that into a formula you can actually use. By definition, $E.T. = \alpha_{mean} - \alpha_{true}$ (a sundial runs ahead when the true Sun's right ascension lags the mean Sun's), and the mean Sun's right ascension is just its mean longitude L. Write $E.T. = (L - \lambda) + (\lambda - \alpha)$ and handle the two pieces separately:

- **Eccentricity:** to first order, the true longitude runs ahead of the mean one by $\lambda - L \approx 2e \sin M$, where M is the mean anomaly. So $L - \lambda \approx -2e \sin M$.
- **Obliquity:** expanding $\tan \alpha = \cos \epsilon \tan \lambda$ for a small tilt effect gives $\lambda - \alpha \approx y \sin 2\lambda$ with $y = \tan^2(\epsilon/2) = 0.0430$.

Those are in radians. Multiply by 229.18 minutes per radian (that's $24 \times 60/2\pi$) and you get

$$\boxed{E.T. \approx -7.66 \sin M + 9.86 \sin 2\lambda \quad \text{minutes.}}$$

The 7.66 is $2e \times 229.18$ and the 9.86 is $y \times 229.18$, which are exactly the $\approx$8 and $\approx$10 minute swings you'd expect. (Differentiate it and you get Figure 17 back: the eccentricity part gives $-7.9 \cos M$ seconds per day, and the obliquity part gives a swing of about $20 \cos 2\lambda$ seconds per day.) To use it you need M and λ for your date. With $d = JD - 2451545.0$:

$$M = 357.529^\circ + 0.98560028^\circ d, \qquad L = 280.460^\circ + 0.98564736^\circ d,$$

$$\lambda = L + 1.915^\circ \sin M + 0.020^\circ \sin 2M.$$

Date (2026)	Formula above	Accurate ephemeris
Feb 11	−14.2 min	−14.2 min
May 14	+3.4 min	+3.7 min
Jul 26	−6.4 min	−6.5 min
Nov 3	+16.4 min	+16.5 min

Table 3: The closed form against a full ephemeris at the four extremes of the year. Over the whole year the two never differ by more than about a quarter of a minute.

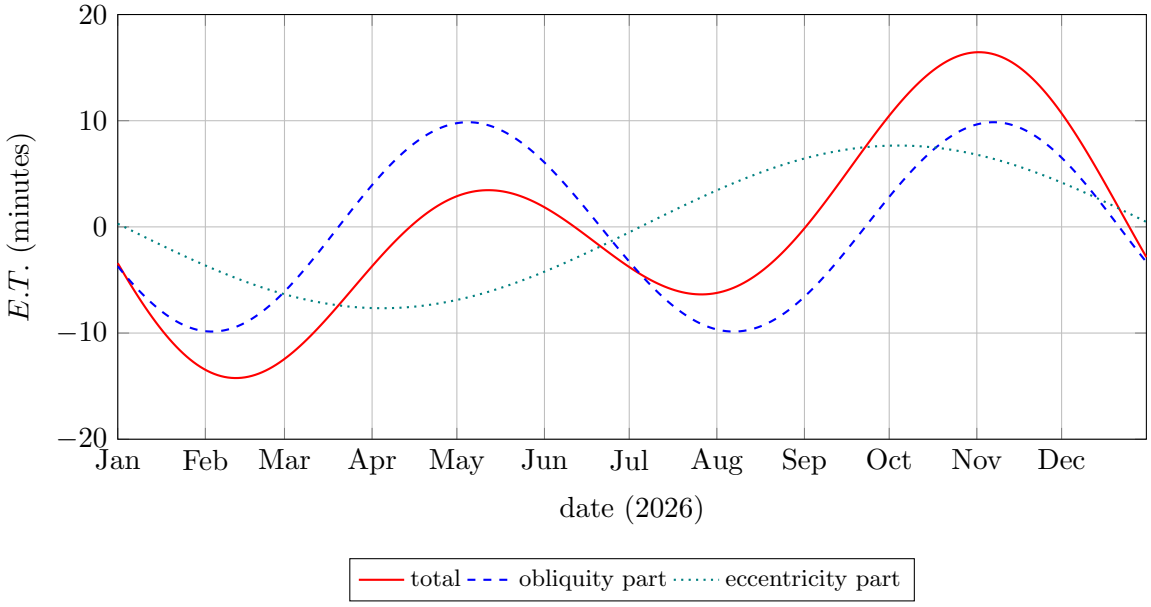


Figure 18: The equation of time (E.T. = apparent − mean) through the year, from the closed form above. The obliquity part (dashed) swings twice a year and the eccentricity part (dotted) once a year. Their sum (solid) peaks at about +16.4 min on November 3 and bottoms out at about −14.2 min on February 11. When the curve is positive, a sundial is ahead of your watch.

17.3 Sidereal Time from a Clock

Now the big one. Section 5 gave you $LST = H + \alpha$, but that's no help if all you know is the date and time. Here's how to get sidereal time straight from the clock. The **Greenwich Mean Sidereal Time** (the sidereal time at longitude 0°) at a given instant is, to a very good approximation,

$$\boxed{GMST = 18.697374558^h + 24.06570982441908^h \times (JD - 2451545.0) \pmod{24^h},}$$

where JD is the Julian Date in UT. (That huge coefficient is just $24^h \times 1.0027379$, because sidereal time runs a bit faster than clock time, which we'll get to in a second.) This is good to well under a second for dates within a few decades of 2000. Then shifting to your own longitude is easy, since sidereal time is just an angle that depends on where you're standing:

$$\boxed{LST = GMST + \frac{\lambda}{15^\circ} \quad (\lambda \text{ positive east}).}$$

Quick sanity check: at J2000.0 itself, $JD - 2451545.0 = 0$, so $GMST = 18.697374558^h = 18^h41^m50.5^s$, which is the well known value.

Sidereal time runs fast. Sidereal days are shorter than solar days, so a sidereal clock ticks faster. To convert an interval:

$$\boxed{\Delta(\text{sidereal}) = 1.00273790935 \times \Delta(\text{solar}).}$$

Over one day that adds up to $86400 \times 0.0027379 = 236.56$ s, so **sidereal time gains $3^m56.56^s$ on the clock every day**, about 2^h per month. (That's the same 1/365.24 effect as before, showing up again.)

Going backwards. If you know the LST you want and need the clock time, flip the formula. Find $GMST_0$ (the GMST at 0^h UT on your date, using the formula above), and then

$$UT = \frac{LST - \lambda/15^\circ - GMST_0}{1.00273790935} \quad \text{(reduce the numerator into } [0^h, 24^h) \text{ first).}$$

17.4 Which Stars Are Up Tonight?

Here's a trick that works with no calculator. From Section 12.3 you know $LT = H_\odot + 12^h$, and from Section 12.2 you know $LST = H_\odot + \alpha_\odot$. Subtract and you get

$$\boxed{LST = \alpha_\odot + (LT - 12^h).}$$

In words: *the right ascension on your meridian is the Sun's right ascension plus how many hours past noon it is.* And you already know the Sun's R.A. at the four cardinal points (Section 15.3): 0^h on March 20, 6^h on June 21, 12^h on September 22, and 18^h on December 21. In between it grows by about 2^h per month. That gets you $\alpha_\odot$ to within about 20 minutes, which is plenty for "what's overhead tonight."

For example, on January 15 the Sun is about 9.9 months past March 20, so $\alpha_\odot \approx 19^h45^m$. At 22:00 local mean time, $LST \approx 19^h45^m + 10^h = 5^h45^m$. Betelgeuse has $\alpha = 5^h55^m$, so Orion is crossing your meridian at 10 pm in mid January, exactly like every winter night you've ever looked up. (The exact answer is 5^h41^m, so the shortcut is off by a few minutes, which is the equation of time sneaking back in.)

17.5 Sample Problems

Problem: Julian Date of a Classic Example

Find the Julian Date of 1987 April 10 at 19:21 UT.

The month is April, so $M = 4 > 2$ and we don't shift anything. $A = \lfloor 1987/100 \rfloor = 19$ and $B = 2 - 19 + \lfloor 19/4 \rfloor = 2 - 19 + 4 = -13$. Then $\lfloor 365.25 \times (1987 + 4716) \rfloor = \lfloor 365.25 \times 6703 \rfloor = 2448270$, and $\lfloor 30.6001 \times 5 \rfloor = 153$. The day with its fraction is $D = 10 + 19.35/24 = 10.80625$. So

$$JD = 2448270 + 153 + 10.80625 - 13 - 1524.5 = 2446896.30625.$$

Problem: When Is Solar Noon in Quezon City?

On November 3, when is the Sun on the meridian in Quezon City ($\lambda = 121.04^\circ$ E)? The Philippines runs on UTC+8, so the standard meridian is 120° E.

On November 3 the equation of time is $E.T. \approx +16.4$ min (Table 3), so the sundial is ahead and the Sun crosses the meridian when *mean* time reads $12^h - 16.4^m = 11{:}43.6$. Quezon City sits $1.04°$ east of the standard meridian, which means its local mean time is $1.04 \times 4 = 4.2$ minutes ahead of the clock, so the clock reads 4.2 minutes *earlier* than the mean time at the moment of transit:

$$11{:}43.6 \text{ (mean time)} \;-\; 4.2 \text{ min} \;=\; 11{:}39.4 \implies \boxed{\text{about 11:39 am}}.$$

Problem: Sidereal Time in Quezon City

Find the local sidereal time in Quezon City ($\lambda = 121.04°$ E) on 2026 June 21 at 12:00 UT (8 pm local time).

The Julian Date is 2461213.0 (it's noon, so no .5), so $JD - 2451545.0 = 9668.0$. Then $GMST = 18.697374558 + 24.06570982441908 \times 9668 = 232685.980^h$. Reduce mod 24^h to get $GMST = 5.97996^h = 5^h58^m47.8^s$. Quezon City's longitude is $121.04°/15° = 8^h04^m09.6^s$, so

$$LST = 5^h58^m47.8^s + 8^h04^m09.6^s = \boxed{14^h02^m57.4^s}.$$

(A full ephemeris routine gives $14^h02^m57.5^s$.)

Problem: When Does Betelgeuse Transit?

Using the Sun shortcut, estimate the local mean time at which Betelgeuse ($\alpha = 5^h55^m$) crosses the meridian on January 15.

A star transits when $LST = \alpha$. Rearranging $LST = \alpha_\odot + (LT - 12^h)$ gives $LT = \alpha - \alpha_\odot + 12^h = 5^h55^m - 19^h45^m + 12^h = -1^h50^m$, and adding 24^h gives $\boxed{22{:}10}$ local mean time. That's ten minutes after the 22:00 example above, which matches Betelgeuse being ten minutes of R.A. past the 5^h45^m on the meridian at that time.

18 Observing Toolkit: When, Where, and How Much

This section is a grab bag of the formulas you actually reach for when a problem says "when does it rise?", "how far apart are they?", "where is it now?", or "how much does the atmosphere mess it up?". Everything here builds on the triangle from Section 10.

18.1 Rise, Transit, and Set of Any Object

You already know the sunrise formula, $\cos H = -\tan\phi\tan\delta$ (Section 7). The cool part is that it has nothing to do with the Sun. It comes from setting $h = 0$ in $\sin h = \sin\delta\sin\phi + \cos\delta\cos\phi\cos H$, so it works for *any* object. And since $LST = H + \alpha$, you get a full schedule for any star:

$$\boxed{LST_{transit} = \alpha, \qquad LST_{rise} = \alpha - H, \qquad LST_{set} = \alpha + H, \qquad \cos H = -\tan\phi\tan\delta.}$$

(Convert H to hours by dividing by $15°$.) The object is above the horizon for a total of $2H$ of sidereal time. If $|\tan\phi\tan\delta| > 1$ there's no solution: the star is either circumpolar or never rises, which is the same condition as Section 6.3. Need the clock time? Use the "going backwards" formula from Section 17.3.

Where does it rise? Set $h = 0$ in $\sin\delta = \sin h \sin\phi + \cos h \cos\phi \cos A$ (Section 12.1.2) and you get

$$\boxed{\cos A_{rise} = \frac{\sin\delta}{\cos\phi}, \qquad A_{set} = 360^\circ - A_{rise}}$$

with azimuth measured from north through east, as usual. Two quick sanity checks: an object on the celestial equator has $\delta = 0$, so $A_{rise} = 90^\circ$. Everything on the celestial equator rises due east and sets due west, for *everybody*, and stays up for exactly 12 hours of sidereal time. And a northern object ($\delta > 0$) has $A_{rise} < 90^\circ$, so it rises north of east, which is why summer Suns rise in the northeast.

Example. Vega ($\alpha = 18^h36^m56^s$, $\delta = +38^\circ47'$) from $\phi = 14.5^\circ$ N: $\cos H = -\tan 14.5^\circ \tan 38.78^\circ = -0.2078$, so $H = 101.99^\circ = 6^h48^m$. Then $LST_{rise} = 18^h37^m - 6^h48^m = 11^h49^m$ and $LST_{set} = 18^h37^m + 6^h48^m = 1^h25^m$, and the azimuth is $\cos A = \sin 38.78^\circ / \cos 14.5^\circ = 0.647$, so $A_{rise} = 49.7^\circ$ (northeast) and $A_{set} = 310.3^\circ$ (northwest). Vega is up for 13.6 sidereal hours. A real ephemeris gets the same azimuths to within a few hundredths of a degree.

Stars and refraction

For the Sun and Moon we used $h_0 = -0.833^\circ$ (Section 16.4). For a point-like star, only the refraction matters, so use $h_0 = -34' = -0.567^\circ$ in $\cos H = (\sin h_0 - \sin\phi \sin\delta)/(\cos\phi\cos\delta)$. It makes every rise a bit earlier and every set a bit later, by a few minutes at low latitudes (and by more at high latitudes, where things skim along the horizon slowly).

Twilight

Twilight is the same formula with a different target altitude. Civil twilight ends when the Sun's center is 6° below the horizon ($h_0 = -6^\circ$), nautical twilight ends at -12°, and astronomical twilight ends at -18°, when the sky is finally properly dark. Plug that h_0 into $\cos H = (\sin h_0 - \sin\phi\sin\delta)/(\cos\phi\cos\delta)$. At the equator on an equinox the Sun drops straight down at a clean 15° per hour, so the three twilights end about 21 minutes, 45 minutes, and 1^h09^m after sunset. If the right side of that equation comes out below -1, the Sun never gets that low. At $\phi = 52^\circ$ N near the June solstice, $\cos H = -1.10$ for $h_0 = -18^\circ$, so there's no astronomical night at all, just a sky that never quite goes fully dark. That's what a "white night" is.

18.2 Angular Separation

You've already used this twice (Mizar and Alcor, the galactic pole), so let's write it down properly. For two objects with coordinates (α_1, δ_1) and (α_2, δ_2), the spherical law of cosines with sides $90^\circ - \delta_1$ and $90^\circ - \delta_2$ gives

$$\boxed{\cos\Theta = \sin\delta_1 \sin\delta_2 + \cos\delta_1 \cos\delta_2 \cos(\alpha_1 - \alpha_2).}$$

It works great, with one trap: when Θ is small, $\cos\Theta$ is crazy close to 1, and rounding wrecks you. For Mizar and Alcor, $\cos\Theta = 0.99999410$. If you round that to six decimals (0.999994), the angle comes out as 0.1985° instead of the true 0.1968°, a nearly 1% error just from rounding. The fix is the **haversine** form, which stays accurate for tiny angles:

$$\boxed{\operatorname{hav}\Theta = \operatorname{hav}(\delta_1 - \delta_2) + \cos\delta_1 \cos\delta_2 \operatorname{hav}(\alpha_1 - \alpha_2), \qquad \operatorname{hav} x = \sin^2\frac{x}{2}, \quad \Theta = 2\sin^{-1}\sqrt{\operatorname{hav}\Theta}.}$$

For Mizar and Alcor, that gives $\operatorname{hav}\Theta = 2.9506 \times 10^{-6}$ and $\Theta = 0.19684^\circ$ with no precision drama. And when the angle is really small, you can skip the sphere entirely and use the same Pythagoras trick as proper motion (Section 14.3): $\Theta \approx \sqrt{(\Delta\delta)^2 + (\Delta\alpha\cos\bar{\delta})^2}$, where $\bar{\delta}$ is the average declination. For Mizar and Alcor, $\Delta\alpha = 19.43'$ and $\cos\bar{\delta} = 0.576$, so $\Delta\alpha\cos\bar{\delta} = 11.19'$, while $\Delta\delta = 3.78'$. Then $\Theta \approx \sqrt{3.78^2 + 11.19^2}' = 11.81' = 0.1968^\circ$. All three methods agree.

18.3 Parallactic Angle

Look back at the triangle NXZ from Section 10.2. We've used the sides and the angles at N and Z, but the angle at the object X is useful too. It's called the **parallactic angle** q: the angle at the object between the direction to the north celestial pole and the direction to the zenith. Basically it tells you how "tilted" the sky's north-south direction is compared to straight up, at that object, right now.

Get it from the law of sines. The side opposite q is $\widehat{ZN} = 90^\circ - \phi$, and the side opposite H is $\widehat{ZX} = 90^\circ - h$, so

$$\frac{\sin q}{\sin(90^\circ - \phi)} = \frac{\sin H}{\sin(90^\circ - h)} \quad \Longrightarrow \quad \boxed{\sin q = \frac{\sin H \cos\phi}{\cos h}.}$$

Heads up: $\sin q$ can't tell q from $180^\circ - q$ (that's the usual sine ambiguity). The version that always gets the right quadrant is

$$\boxed{\tan q = \frac{\sin H}{\tan\phi\cos\delta - \sin\delta\cos H},}$$

and you should evaluate it with atan2 (numerator, denominator). The sign of q follows the sign of H: positive in the west, negative in the east, and $q = 0$ or 180° when the object is on the meridian.

Example. A star at $\delta = +20^\circ$ seen from $\phi = 45^\circ$ N at $H = +40^\circ$ (west of the meridian): $\tan q = \sin 40^\circ / (\tan 45^\circ \cos 20^\circ - \sin 20^\circ \cos 40^\circ) = 0.6428/0.6778$, so $q = 43.5^\circ$.

Why care? Two big reasons. On an alt-az telescope, the field of view rotates by q as the object crosses the sky, which is why long exposures need a field derotator. And atmospheric refraction smears starlight along the vertical, so q tells you which way the smear points relative to north.

18.4 Precession of Coordinates

Here's something that catches people off guard: the R.A. and declination in a catalogue are for an *epoch* (usually J2000.0), and the stars aren't there tonight. We saw in Section 15.3.4 that precession slides the equinox westward along the ecliptic at $p \approx 50.3''$ per year, so every star's ecliptic longitude grows by p per year while its ecliptic latitude stays put. Let's turn that into changes in α and δ.

Take the ecliptic to equatorial relations with β held fixed and λ increasing by $d\lambda$. Differentiating $\sin\delta = \cos\epsilon\sin\beta + \sin\epsilon\cos\beta\sin\lambda$ and using $\cos\delta\cos\alpha = \cos\beta\cos\lambda$ gives $d\delta = \sin\epsilon\cos\alpha\, d\lambda$, and the same kind of differentiation on the formula for α gives $d\alpha = (\cos\epsilon + \sin\epsilon\sin\alpha\tan\delta)\, d\lambda$. Put $d\lambda = p\, dt$:

$$\boxed{\frac{d\alpha}{dt} = m + n\sin\alpha\tan\delta, \qquad \frac{d\delta}{dt} = n\cos\alpha,}$$

$$m = p\cos\epsilon \approx 3.075 \text{ s/yr } (46.1''/\text{yr}), \qquad n = p\sin\epsilon \approx 1.336 \text{ s/yr } (20.04''/\text{yr}).$$

(The textbook values of m and n differ from $p\cos\epsilon$ and $p\sin\epsilon$ in the third digit, because the ecliptic plane itself drifts a tiny bit.) Notice that the $\tan\delta$ in the α formula means R.A. changes fastest for stars near the poles.

Example. Vega, from J2000.0 to 2026 June 21 ($t = 26.47$ yr). Here $\sin\alpha\tan\delta = -0.793$ and $\cos\alpha = 0.161$, so

$$\Delta\alpha = (3.075 - 1.336 \times 0.793) \times 26.47 = +53.3 \text{ s}, \qquad \Delta\delta = 20.04'' \times 0.161 \times 26.47 = +85''.$$

That moves Vega from $(18^h36^m56.3^s,\ +38°47'01'')$ to $(18^h37^m49.6^s,\ +38°48'26'')$, which matches a rigorous precession calculation to a tenth of a second of time and about an arcsecond. Pretty good for a two-line formula.

Warning. This first-order formula falls apart near the celestial poles, where $\tan\delta$ blows up. For Polaris ($\delta = 89.26°$), the formula's R.A. change over the same 26.8 years is off by about 15%. Near a pole, use the full rotation matrices from Section 15.5 instead. (Also, none of this includes *proper motion*, which is each star's own drift from Section 14.3.)

18.5 Refraction and Airmass

Two atmosphere formulas that show up a lot. First, **refraction**. Starlight bends as it enters the atmosphere, so everything looks higher than it truly is. A good formula (Bennett's) for the refraction R, in arcminutes, in terms of the *apparent* altitude h in degrees, for standard conditions (1010 hPa, 10°C), is

$$\boxed{R = \cot\left(h + \frac{7.31}{h + 4.4}\right) \text{ arcminutes}}$$

and then the true altitude is $h_{true} = h - R/60$ (in degrees). It's accurate to about $0.1'$ over most of the sky:

h	90°	60°	45°	30°	20°	10°	5°	0°
R	$0'$	$0.57'$	$0.99'$	$1.7'$	$2.7'$	$5.4'$	$9.9'$	$34.5'$

Table 4: Refraction against apparent altitude. Remember "about $1'$ at 45° and about $34'$ at the horizon."

That $34.5'$ at the horizon is the number behind the $34'$ in Section 16.4. It also explains why the setting Sun looks squashed: the bottom edge sits at the horizon and gets lifted by about $34'$, but the top edge is about half a degree higher and only gets lifted by about $28'$, so the Sun's vertical size shrinks by roughly $6'$ (nearly 20% of its diameter) while its width stays the same.

Second, **airmass**, the amount of atmosphere you're looking through, normalized to 1 at the zenith. If the atmosphere were a flat slab (the same assumption behind the cosecant law in Section 15.4), looking at zenith distance z would take you through $1/\cos z$ times as much air:

$$\boxed{X \approx \sec z = \frac{1}{\sin h}.}$$

This is accurate to about 1% above $h = 20°$ and then falls apart, because the Earth is round and the real airmass at the horizon is only about 38, not infinity. A formula that works all the way down (Kasten and Young) is

$$X = \frac{1}{\sin h + 0.50572\,(h + 6.07995°)^{-1.6364}}.$$

For example, at $h = 10°$ the slab formula gives 5.76 and Kasten and Young gives 5.59. Why you care: starlight dims by about k magnitudes per airmass, so $m_{observed} = m_{true} + kX$, where k is the extinction coefficient (typically a few tenths of a magnitude at a decent site). Observing at airmass 2 instead of 1 costs you a few tenths of a magnitude, which is why astronomers wait for their targets to climb.

18.6 Synodic Periods

You've secretly already used this idea. The solar day came from $1/T_{sol} = 1/T_{rot} - 1/P$, which is the time it takes for the Earth's spin to "lap" the Sun's apparent motion. That same math gives the **synodic period**: the time between successive alignments of two bodies orbiting the Sun in the same direction, like the time between one opposition of Mars and the next. Two bodies with sidereal periods $P_1 < P_2$ have angular speeds $2\pi/P_1$ and $2\pi/P_2$, and the faster one has to gain one full lap on the slower one:

$$\boxed{\frac{1}{S} = \left|\frac{1}{P_1} - \frac{1}{P_2}\right|.}$$

For planets, one of the two periods is Earth's year:

	P (sidereal)	S (synodic)	S in years
Mercury	87.969 d	115.88 d	0.317
Venus	224.701 d	583.92 d	1.599
Mars	686.980 d	779.94 d	2.135
Jupiter	4332.589 d	398.88 d	1.092
Saturn	10759.22 d	378.09 d	1.035

Notice the pattern: the farther out the planet, the more it just sits there while you lap it, so S gets closer to one Earth year. The Moon works the same way. Its sidereal month is 27.3217 d, and the Sun is the thing it has to catch up with (Earth's orbit takes the role of the second period), so $1/S = 1/27.3217 - 1/365.2564$ gives $S = 29.531$ d, the lunar phase cycle. A fun coincidence: five Venus synodic periods are $5 \times 583.92 = 2919.6$ d, while eight years is 2921.9 d, so Venus returns to nearly the same spot in the sky every eight years, tracing a perfect five petal pattern.

18.7 Sample Problems

Problem: When Does Vega Rise in Quezon City?

Find the rise and set times (in UT and in Philippine time, UTC+8) of Vega on 2026 June 21 from Quezon City ($\phi = 14.5°$ N, $\lambda = 121.04°$ E). Vega's J2000 coordinates are $(18^h36^m56.3^s,\ +38°47'01.3'')$. Ignore refraction.

Step 1, precess to the date. From Section 18.4 with $t = 26.47$ yr, Vega's coordinates of date are $\alpha = 18^h37^m49.6^s$ and $\delta = +38°48'26''$. (Skipping this step would put your answer off by almost a minute.)

Step 2, the hour angle. $\cos H = -\tan 14.5° \tan 38.807° = -0.2080$, so $H = 102.00° = 6^h48^m01^s$.

Step 3, rise and set in LST. $LST_{rise} = 18^h37^m49.6^s - 6^h48^m01^s = 11^h49^m48.6^s$ and $LST_{set} = 18^h37^m49.6^s + 6^h48^m01^s = 25^h25^m50.6^s = 1^h25^m50.6^s$.

Step 4, convert to the clock. The GMST at 0^h UT on 2026 June 21 is $GMST_0 = 17^h56^m49.6^s$, and $\lambda/15^\circ = 8^h04^m09.6^s$. For the rise: $11^h49^m48.6^s - 8^h04^m09.6^s - 17^h56^m49.6^s = -14^h11^m10.6^s$, so add 24^h to get $9^h48^m49.4^s$, then divide by 1.00273790935:

$$UT_{rise} = 9^h47^m13^s \quad \Longrightarrow \quad \boxed{\text{17:47 Philippine time}}.$$

For the set:

$$1^h25^m50.6^s - 8^h04^m09.6^s - 17^h56^m49.6^s = -24^h35^m08.6^s,$$

so add 48^h to get $23^h24^m51.4^s$, divide by the same factor, and you get

$$UT_{set} = 23^h21^m01^s \quad \Longrightarrow \quad \boxed{\text{07:21 Philippine time}}.$$

That's the next morning, of course. A full ephemeris calculation agrees to within a minute. The rise azimuth is $\cos A = \sin 38.807^\circ / \cos 14.5^\circ$, so $A_{rise} = 49.7^\circ$.

Problem: Three Ways to Split Mizar and Alcor

Using $\alpha_M = 13^h26^m16^s$, $\delta_M = 54^\circ51'33.2''$, $\alpha_A = 13^h24^m58.3^s$, $\delta_A = 54^\circ47'46.5''$, find the separation using the cosine form, the haversine form, and the flat sky approximation.

The cosine form gives $\cos\Theta = 0.99999410$, so $\Theta = 0.19684^\circ$ (if you carry at least eight digits). The haversine form gives $\mathrm{hav}\,\Theta = 3.02 \times 10^{-7} + 2.65 \times 10^{-6} = 2.9506 \times 10^{-6}$, and $\Theta = 2\sin^{-1}\sqrt{2.9506 \times 10^{-6}} = 0.19684^\circ$. The flat sky approximation, with $\Delta\alpha\cos\bar{\delta} = 11.19'$ and $\Delta\delta = 3.78'$, gives $\Theta \approx 11.81' = 0.1968^\circ$. They all agree, but only the last two survive sloppy rounding.

Problem: Synodic Month from the Sidereal Month

The Moon orbits Earth in 27.3217 days (sidereal month). Earth's year is 365.2564 days. Find the synodic month.

The Moon has to catch up with the Sun's apparent motion, which advances by one Earth orbit per year. So $1/S = 1/27.3217 - 1/365.2564 = 0.036602 - 0.002738 = 0.033864$ d^{-1}, which gives $S = 29.531$ days. The 2 day gap between the sidereal and synodic months is exactly the extra distance the Moon has to go because Earth moved along its orbit in the meantime.

19 Formula Cheat Sheet

Everything you need in one place, with section numbers so you can jump back to the derivation. Conventions throughout: azimuth A is measured from north through east, hour angle H is measured westward, longitude λ is positive east, and ϕ is latitude.

Sky geometry	**Formula**	**Sec.**
Local sidereal time	$LST = H + \alpha$	5
Max and min altitude	$h_u = 90^\circ - \lvert\phi - \delta\rvert, \quad h_l = \lvert\phi + \delta\rvert - 90^\circ$	6.2
Circumpolar, never rises	circumpolar if $\lvert\phi + \delta\rvert > 90^\circ$; never rises if $\lvert\phi - \delta\rvert > 90^\circ$	6.3
Altitude	$\sin h = \sin\delta \sin\phi + \cos\delta \cos\phi \cos H$	12.1.1
Declination from alt-az	$\sin\delta = \sin h \sin\phi + \cos h \cos\phi \cos A$	12.1.2
Rise and set hour angle	$\cos H = -\tan\phi \tan\delta$	7, 18.1
Rise azimuth	$\cos A_{rise} = \sin\delta / \cos\phi, \quad A_{set} = 360^\circ - A_{rise}$	18.1
Dip of the horizon	$\cos\Theta = R/(R+h)$	8
Spherical triangles	$\cos a = \cos b \cos c + \sin b \sin c \cos A, \quad \dfrac{\sin A}{\sin a} = \dfrac{\sin B}{\sin b}$	11
Spherical excess	$A_{area} = ER^2, \quad E = A + B + C - 180^\circ$ (radians)	11.4
Sun's declination	$\sin\delta_\odot = \sin\epsilon \sin\lambda_\odot, \quad \tan\alpha_\odot = \cos\epsilon \tan\lambda_\odot$	13.3
Equatorial to ecliptic	$\hat{r}_{ecl} = R_x(\epsilon)\, \hat{r}_{eq}, \quad \epsilon = 23.4393^\circ$	15.5
Equatorial to galactic	$\hat{r}_{gal} = \mathbf{T}\, \hat{r}_{eq}, \quad$ NGP at $(12^h51^m26^s, +27.13^\circ)$	15.5
Cosecant law (dust)	$A(b) \approx A_\perp / \sin\lvert b\rvert$	15.4

Time	**Formula**	**Sec.**
Julian Date	$JD = \lfloor 365.25(Y + 4716) \rfloor + \lfloor 30.6001(M + 1) \rfloor + D + B - 1524.5$	17.1
Greenwich sidereal time	$GMST = 18.6974^h + 24.0657098^h\,(JD - 2451545.0)$	17.3
Local sidereal time	$LST = GMST + \lambda/15^\circ$	17.3
Sidereal and solar intervals	$\Delta(\text{sidereal}) = 1.00273791\,\Delta(\text{solar})$	17.3
Which stars are up	$LST = \alpha_\odot + (LT - 12^h)$	17.4
Zone time	$LMT - (\text{zone time}) = 4\text{ min} \times (\lambda - \lambda_{std})/1^\circ$	17.2
Equation of time	$E.T. \approx -7.66 \sin M + 9.86 \sin 2\lambda$ min (apparent − mean)	17.2
Solar day on any planet	$1/T_{sol} = 1/T_{rot} \mp 1/P$ (prograde −, retrograde +)	16.2
Tropical year	$1/T_{trop} = 1/T_{sid} + 1/T_{prec}$	16.3
Daylight length	$D = H/7.5^\circ$ hours, $\quad \cos H = -\tan\phi \tan\delta$	16.4
With refraction and size	$\cos H = (\sin h_0 - \sin\phi \sin\delta)/(\cos\phi \cos\delta), \quad h_0 = -0.833^\circ$	16.4
Solar day length	$\Delta T \approx 235.91\text{ s} \times [\cos\epsilon\,(1 + 2e\cos M)/\cos^2\delta_\odot - 1]$	16.5
Synodic period	$1/S = \lvert 1/P_1 - 1/P_2 \rvert$	18.6

Observing	**Formula**	**Sec.**
Rise, transit, set (LST)	$LST_{rise} = \alpha - H, \quad LST_{transit} = \alpha, \quad LST_{set} = \alpha + H$	18.1
Angular separation	$\text{hav}\,\Theta = \text{hav}\,\Delta\delta + \cos\delta_1 \cos\delta_2\,\text{hav}\,\Delta\alpha$	18.2
Parallactic angle	$\tan q = \sin H/(\tan\phi\cos\delta - \sin\delta\cos H)$	18.3
Precession	$\dot\alpha = m + n\sin\alpha\tan\delta, \quad \dot\delta = n\cos\alpha \quad (m = 3.075$ s/yr, $n = 20.04''$/yr)	18.4
Proper motion	$u = \sqrt{\dot\delta^{\,2} + (\dot\alpha\cos\delta)^2}$	14.3
Refraction	$R = \cot\big(h + 7.31/(h + 4.4)\big)$ arcmin	18.5
Airmass	$X \approx \sec z = 1/\sin h$	18.5